



Quantitative profiling of pseudouridylation dynamics in native RNAs with nanopore sequencing

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Nanopore RNA sequencing shows promise as a method for discriminating and identifying different RNA modifications in native RNA. Expanding on the ability of nanopore sequencing to detect N⁶-methyladenosine, we show that other modifications, in particular pseudouridine (Ψ) and 2'-O-methylation (Nm), also result in characteristic base-calling 'error' signatures in the nanopore data. Focusing on Ψ modification sites, we detected known and uncovered previously unreported Ψ sites in mRNAs, non-coding RNAs and rRNAs, including a Pus4-dependent Ψ modification in yeast mitochondrial rRNA. To explore the dynamics of pseudouridylation, we treated yeast cells with oxidative, cold and heat stresses and detected heat-sensitive Ψ-modified sites in small nuclear RNAs, small nucleolar RNAs and mRNAs. Finally, we developed a software, nanoRMS, that estimates per-site modification stoichiometries by identifying single-molecule reads with altered current intensity and trace profiles. This work demonstrates that Nm and Ψ RNA modifications can be detected in cellular RNAs and that their modification stoichiometry can be quantified by nanopore sequencing of native RNA.

RNA modifications are chemical moieties that decorate RNA molecules, expanding their lexicon. By coupling antibody immunoprecipitation or chemical probing with next-generation sequencing (NGS), transcriptome-wide maps of several RNA modifications have been constructed, including N⁶-methyladenosine (m⁶A)^{1,2}, pseudouridine (Ψ)^{3–6}, N⁵-methylcytosine (m⁵C)^{7,8}, 5-hydroxymethylcytosine (hm⁵C)⁹, N¹-methyladenosine (m¹A)^{10,11}, N³-methylcytosine (m³C)¹², N⁴-acetylcytosine (ac⁴C)^{13,14} and N⁷-methylguanosine (m⁷G)^{15,16}. These studies revealed that RNA modifications play a pivotal role in many cellular processes, including regulation of cellular fate¹⁷, sex determination¹⁸ and cellular differentiation¹⁹.

Despite these advances, a fundamental challenge in the field is the lack of a generic approach for mapping diverse RNA modification types simultaneously^{20–23}. Currently, customized protocols must be optimized for each RNA modification type, leading to experimental designs in which the RNA modification type to be studied is chosen beforehand, limiting our ability to characterize the plasticity of the epitranscriptome in a systematic and unbiased manner. Moreover, even in those cases where a selective antibody or chemical is available, NGS-based methods are often not quantitative, have high false-positive rates²¹, are inconsistent when using distinct antibodies²⁴, are unable to produce maps for highly repetitive regions, cannot provide information regarding the co-occurrence of distant modifications in same transcripts, do not provide isoform-specific information and require multiple ligations steps and extensive polymerase chain reaction amplification during the library preparation, introducing undesired biases in the sequencing data²⁵.

A promising alternative to NGS-based technologies is the direct RNA sequencing platform developed by Oxford Nanopore Technologies (ONT), which has the potential to detect virtually any given RNA modification present in native RNA molecules^{20,26,27}. Several studies showed that RNA modifications can be identified using this technology^{28–30}; however, most efforts have so far been limited to the detection of m⁶A modifications^{28–34}, and it is largely unknown whether other RNA modifications might be distinguishable from their unmodified counterparts. Thus, a systematic approach that can map and quantify diverse RNA modifications simultaneously is currently missing.

In this study, we examined the *Saccharomyces cerevisiae* coding and non-coding transcriptome at single-molecule resolution using native RNA nanopore sequencing. We found that most RNA modifications cause systematic base-calling errors and that the signature of these base-calling 'errors' can be used to identify the underlying RNA modification type. For example, we found that Ψ typically appears in the form of U-to-C mismatches, whereas m⁵C modifications appear in the form of insertions. We then exploited the identified signatures to de novo predict RNA modifications in rRNAs, identifying two previously unreported Ψ modifications in mitochondrial rRNA, which we confirmed using CMC probing coupled to nanopore sequencing (nanoCMC-seq). We demonstrate that one of these Ψ modifications (15s:Ψ854) is placed by the enzyme Pus4, which was previously thought to pseudouridylate only mRNAs and transfer RNAs (tRNAs)⁴. Moreover, we show that, once RNA modifications have been accurately predicted using base-calling 'errors', the stoichiometry of a given Ψ- or Nm-modified site can be estimated by

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clustering per-read features (current intensities and trace) of the modified regions.

We then explored the dynamics of RNA modifications present in non-coding RNAs (ncRNAs). It has been proposed that differential rRNA modifications might constitute a source of ribosomal heterogeneity^{35–37}. Indeed, previous studies showed that temperature changes affect rRNA pseudouridylation levels, suggesting that cells might be able to generate compositionally distinct ribosomes in response to environmental cues^{4,38,39}. Similarly, alterations in the stoichiometry of 2'-O-methylation (Am, Cm, Gm and Um)^{40–42} and pseudouridylation (Ψ)^{35–37} have been found to affect translation initiation of mRNAs containing internal ribosome entry sites^{43,44}. Here we re-examine this question using direct RNA sequencing and characterize the RNA modification dynamics in rRNAs, small nuclear RNAs (snRNAs) and small nucleolar RNAs (snoRNAs) upon a battery of environmental cues, translational repertoires and genetic strains. Contrary to expectations, we found that none of the environmental stresses tested leads to significant changes in the ribosomal epitranscriptome. By contrast, our method does recapitulate previously reported heat-dependent Ψ snRNA modifications, as well as identifies previously unreported heat-sensitive sites in snRNAs and snoRNAs.

Finally, we developed an algorithm, nanoRMS, that can predict Ψ RNA modifications *de novo* and estimate the stoichiometry of modification in both highly and lowly modified Ψ and Nm sites across diverse types of RNA molecules, including rRNAs, sn/snoRNAs and mRNAs. Our approach recapitulates known Pus1-dependent, Pus4-dependent and heat stress-dependent mRNA sites as well as reveals Ψ mRNA sites that had not been previously reported. Altogether, our work establishes a framework for the study of RNA modification dynamics using direct RNA nanopore sequencing, opening avenues to study the plasticity of the epitranscriptome at single-molecule resolution.

Results

RNA modification detection depends on base-calling and mapping algorithms. Previous studies have shown that m⁶A RNA modifications can be detected in the form of non-random base-calling 'errors' in direct RNA sequencing datasets^{29–33}. However, it is unclear how these 'errors' might vary with the choice of base-calling and mapping algorithms and, consequently, affect the ability to detect RNA modifications. In this study, we compared the performance of commonly used base-calling and mapping algorithms on *in vitro* transcribed RNA sequences that contained all possible combinations of 5-mers, referred to as 'curlcakes'²⁹, that included: (1) unmodified nucleosides (UNM), (2) m⁶A, (3) Ψ , (4) m⁵C and (5) hm⁵C (Fig. 1a). In addition, a sixth dataset containing unmodified short RNAs (UNM-S), with median length of 200 nucleotides, was included in the analysis to assess the effect of input sequence length in base-calling (Methods). Each dataset was base-called with

two distinct algorithms (Albacore and Guppy), and two different versions for each of them were used: (1) Albacore version 2.1.7 (AL 2.1.7); (2) its latest version, Albacore 2.3.4 (AL 2.3.4); (3) Guppy 2.3.1 (GU 2.3.1); and (4) a more recent version of the latter base-caller, Guppy 3.0.3 (GU 3.0.3), which employs a flip-flop algorithm. We found that the latest version of Albacore (2.3.4) base-called 100% of sequenced reads in all six datasets, whereas its previous version did not (average of 90.8%) (Fig. 1b). By contrast, both versions of Guppy (2.3.1 and 3.0.3) produced similar results in terms of percentage of base-called reads (99.96% and 100%, respectively) (Supplementary Table 1).

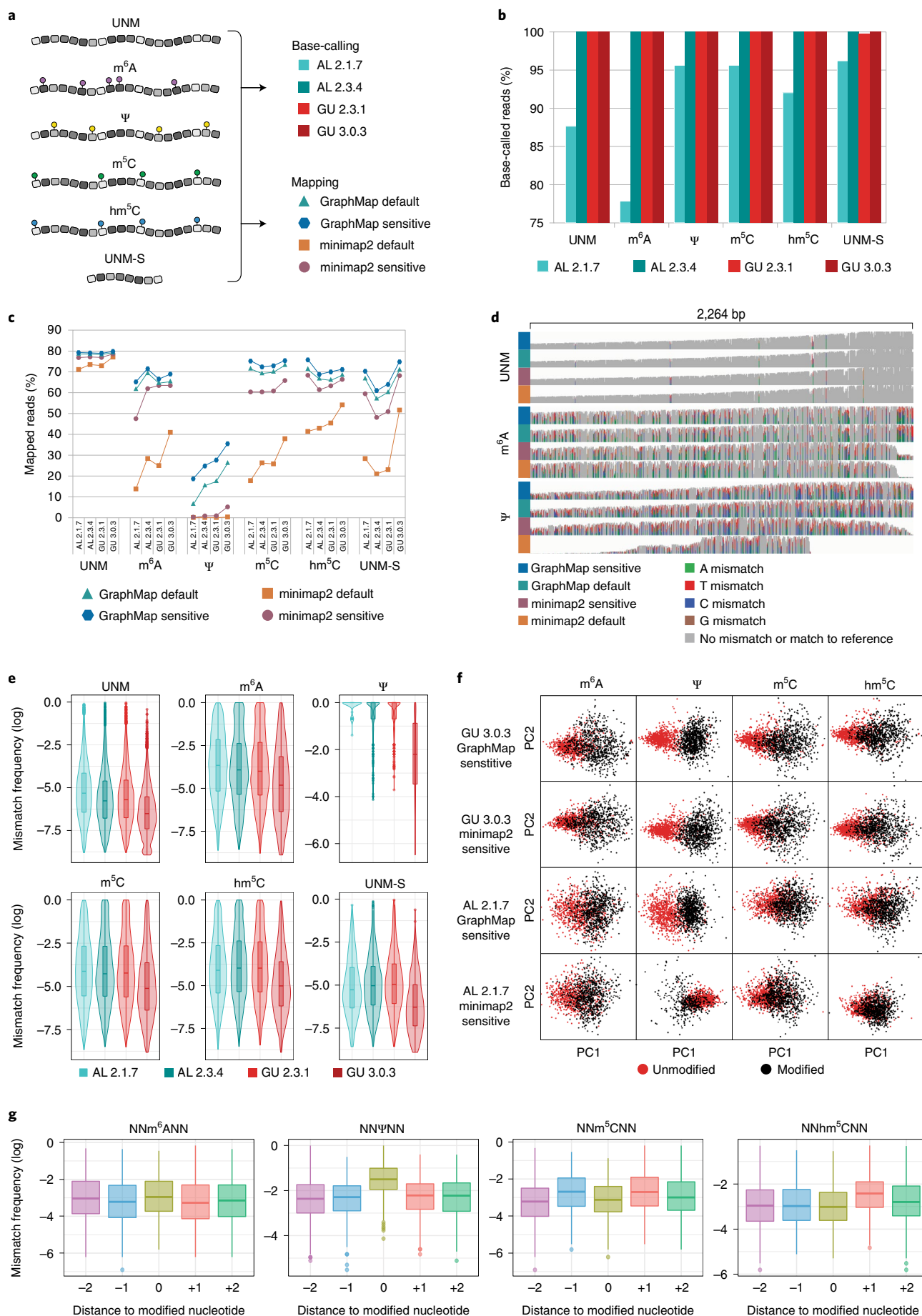
We then assessed whether the choice of mapper might affect the ability to detect RNA modifications. To this end, we employed two commonly used long-read mappers, minimap2 (ref. ⁴⁵) and GraphMap⁴⁶, using either 'default' or 'sensitive' parameter settings (see Methods). Notably, the choice of both mapper and parameters used severely affected the number of mapped reads (Fig. 1c and Supplementary Table 1). The most extreme case was observed with the Ψ -modified dataset, where minimap2 was unable to map the majority of the reads (0–0.3% mapped reads) (Fig. 1c,d and Supplementary Fig. 1a). By contrast, GraphMap 'sensitive' was able to map 35.5% of Ψ -modified base-called reads, with only a minor loss in accuracy (3%) (Supplementary Fig. 1b and Supplementary Table 2), proving to be a more appropriate choice for highly modified datasets.

Base-calling 'error' signatures can be used to predict RNA modification type. Although base-calling 'errors' can be used to identify m⁶A RNA modified sites^{29,30,32}, whether this approach is applicable for the detection of other RNA modifications, and whether these signatures could be employed to distinguish among distinct RNA modification types, are largely unknown. To this end, we systematically characterized the base-calling errors caused by the presence of m⁶A, Ψ , m⁵C and hm⁵C. We found that, regardless of the base-caller and mapper settings used, modified RNA sequences presented decreased quality scores (Supplementary Fig. 1c–e) and higher mismatch frequencies (Fig. 1e), and that these differences were more prominent in Ψ -modified datasets. Principal component analysis (PCA) of base-calling 'errors' of each modified dataset (m⁶A, Ψ , m⁵C and hm⁵C)—relative to unmodified—showed that this difference was greatest in Ψ -modified datasets (Fig. 1f) and maximized in datasets that were base-called with GU 3.0.3. Thus, we found that all four RNA modifications can be detected in direct RNA sequencing data; however, their detection is severely affected by the choice of both base-calling and mapping algorithms, as well as on the RNA modification type.

We then examined whether the base-called 'errors' observed in modified and unmodified datasets occurred in the modified position. We found that both m⁶A and Ψ modifications led to increased mismatch frequencies at the modified site (Fig. 1g), mainly in

Fig. 1 | Systematic analysis of base-calling and mapping algorithms for the detection of RNA modifications in direct RNA sequencing datasets.

a, Overview of the synthetic constructs used to benchmark the algorithms, which included both unmodified (UNM and UNM-S) and modified (m⁶A, m⁵C, hm⁵C and Ψ) sequences. For each dataset, we performed (i) comparison of base-calling algorithms, (ii) comparison of mapping algorithms, (iii) detection of RNA modifications using base-called features and (iv) comparative analysis of features to distinguish similar RNA modifications. **b**, Bar plots comparing the percentage of base-called reads using four different base-calling algorithms in six different unmodified and modified datasets. **c**, Relative proportion of base-called and mapped reads using all possible combinations (16) of base-callers and mappers included in this study, for each of the six datasets analyzed. **d**, IGV snapshots illustrating the differences in mapping for three distinct datasets: UNM, m⁶A-modified and Ψ -modified when base-called with GU 3.0.3. Positions with mismatch frequencies greater than 0.1 are colored; gray represents match to reference. **e**, Comparison of global mismatch frequencies using different base-calling algorithms, for the six datasets analyzed. Box, first to last quartiles; whiskers, 1.5× interquartile range; center line, median; points, outliers; violin, distribution of density. **f**, PCA using as input the base-calling error features of quality, mismatch frequency and deletion frequency in positions −2, −1, 0, 1 and 2, for all datasets base-called with GU 3.0.3 and AL 2.1.7 and mapped with GraphMap and minimap2 on sensitive settings. Only *k*-mers that contained a modification at position 0 and no other modifications in the 5-mer were included in the analysis, and the equivalent set of unmodified *k*-mers was used as a control. **g**, Mismatch frequency of each position of the 5-mers centered in the modified position (position 0). Box, first to last quartiles; whiskers, 1.5× interquartile range; center line, median; points, outliers. See also Supplementary Fig. 1.



the form of U-to-C mismatches in the case of Ψ modifications (Supplementary Fig. 1f). By contrast, m^5C and hm^5C modifications did not appear in the form of increased mismatch frequencies at the modified site; rather, these modifications appeared in the form of increased mismatch frequencies in the neighboring residues (position -1 and $+1$ in the case of m^5C modifications; position $+1$ in hm^5C) (Fig. 1g). Moreover, the base-called ‘error’ signatures of m^5C and hm^5C were also dependent on the sequence context (Supplementary Fig. 1g). Altogether, we found that all four RNA modifications studied (m^6A , m^5C , hm^5C and Ψ) cause base-calling ‘errors’ and that these ‘errors’ follow specific patterns that depend on the RNA modification type.

Ψ modifications can be detected as U-to-C mismatches. We then examined whether the results obtained using in vitro transcribed constructs would be applicable to in vivo RNA sequences. To this end, total RNA from *S. cerevisiae* was prepared for direct RNA nanopore sequencing (see Methods). Visual inspection of the mapped reads revealed a high proportion of base-calling errors present in 25s and 18s rRNAs, as could be expected from sequences that are highly enriched in RNA modifications (Fig. 2a). By contrast, 5s and 5.8s rRNAs did not show such base-calling errors, in agreement with their low level of modification.

Next, we systematically analyzed base-called features (mismatch, deletion, insertion and per-base qualities) of rRNA modified sites relative to unmodified ones (Fig. 2b) and found that all rRNA modification types consistently led to decreased per-base qualities at modified sites, suggesting that per-base qualities can be employed to identify RNA modifications but not the underlying RNA modification type. We found that Ψ modifications caused significant variations in mismatch frequencies, in agreement with our observations using in vitro constructs. By contrast, other RNA modifications, such as 2'-O-methylcytidine (Cm) or m^5C , did not appear in the form of increased mismatch frequencies at modified sites but, rather, in the form of increased insertions. In addition, Ψ modifications typically appeared in the form of U-to-C mismatches (Fig. 2c and Supplementary Fig. 2), in agreement with our in vitro observations, whereas other RNA modifications, such as 2'-O-methyladenosine (Am), did not cause mismatches with unique directionality. Thus, we conclude that distinct rRNA modification types can be detected in the form altered base-called features in vivo and that their base-calling ‘error’ signature is dependent on the RNA modification type.

To confirm that the detected signal (U-to-C mismatches) in Ψ positions was caused by the presence of the Ψ modification, we compared rRNA modification profiles from wild-type *S. cerevisiae* to those from snoRNA knockout strains (snR3, snR34 and snR36) (Fig. 3a and Supplementary Table 3). Our results show that changes in rRNA modification profiles were consistently and exclusively observed in those Ψ positions reported as targets of each snoRNA. Moreover, the remaining Ψ -modified positions were not significantly altered by the lack of Ψ modifications guided by snR3, snR34 or snR36 (Fig. 3b).

We then sequenced three additional *S. cerevisiae* strains depleted of snoRNAs (snR60, snR61 and snR62, respectively) guiding

2'-O-methylation (Nm) at specific positions (Supplementary Table 3). In contrast to Ψ modifications, we found that 2'-O-methylations often caused increased errors not only at the modified position but also at neighboring positions (Fig. 3c and Supplementary Fig. 3a). These errors disappeared in the knockout strain, confirming that neighboring base-calling errors were indeed caused by the 2'-O-methylation (Fig. 3c). On the other hand, although Ψ modifications mainly affected mismatch frequency, we observed that Nm modifications often affected several base-called ‘error’ features (mismatch, insertion and deletion frequency) (Supplementary Fig. 3b). Thus, we reasoned that combining all three features might improve the signal-to-noise ratio for the detection of 2'-O-methylated sites and found that the combination of features led to improved detection of Nm-modified sites (Fig. 3d).

Current intensity variations cannot accurately predict the modified site. We then wondered whether Ψ and Nm sites would also be detected at the level of current intensity changes. We observed that certain Ψ - and Nm-modified sites, such as 25s: Ψ 2129 or 25s:Am1133, showed drastic alterations in their current intensity values in the snoRNA-depleted strain, whereas no significant alteration was observed in control sites (Fig. 3e,f). However, in other sites, the distribution of current intensities did not significantly change in the knockout strain (18s: Ψ 1187; Fig. 3e, lower panel) or did not differ in their mean (25s: Ψ 2133; Supplementary Fig. 4a).

We hypothesized that deviations in current intensity alterations might not always be maximal in the modified site but might sometimes appear in neighboring sites. To test this, we examined the difference in current intensity values along the rRNA molecules for each wild-type-knockout pair (Fig. 4a and Supplementary Fig. 4b). However, the highest deviations in current intensity were often not observed at the modified sites (Fig. 4a, lower panel). From all six Ψ sites that were depleted in the three knockout strains studied, only two of them (25s: Ψ 2826 and 25s: Ψ 2880) showed a maximal deviation in current intensity in the modified site (Fig. 4b and Supplementary Fig. 4c). Similarly, depletion of Nm sites led to changes in current intensity values, but the largest deviations were not observed at the modified site (Supplementary Fig. 4c). Thus, we conclude that current intensity-based methods can detect both Ψ and Nm RNA modifications; however, base-calling errors are a better choice to achieve single-nucleotide resolution, at least in the case of Ψ RNA modifications.

Detection of Ψ and Nm modifications in individual reads. Direct RNA sequencing produces current intensity measurements for each individual native RNA molecule. Thus, modification stoichiometries can be, in principle, estimated by identifying the proportion of reads with altered current intensity at a given site. To this end, we first examined the per-read current intensity values of wild-type and knockout strains at the Ψ - and Nm-depleted sites. Despite the significant variability of current intensities across reads, we were able to observe robust differences in current intensities across strains at the depleted RNA modified sites at the per-read level (Fig. 4c, upper panel). As a control, we performed the same analysis in Ψ sites unaffected by snoRNA depletion, finding no significant

Fig. 2 | RNA modifications can be detected in yeast rRNA in the form of base-calling errors, and each RNA modification type shows a distinct ‘error’ signature. a, IGV snapshots of yeast ribosomal subunits 5s, 5.8s, 18s and 25s. Known modification sites are indicated below each snapshot; nucleotides with mismatch frequencies greater than 0.1 are colored; and gray represents match to reference or no mismatch. **b**, Comparison of base-calling features (base quality, mismatch, deletion and insertion frequency) from distinct RNA modification types present in yeast rRNA. The most descriptive base-calling error per modification is outlined in red. Only RNA modification sites without additional neighboring RNA modifications in the 5-mer were included in the analysis: Ψ ($n=37$), Gm ($n=8$), m^1A ($n=2$), Am ($n=14$), m^5C ($n=2$), Cm ($n=8$), ac^4C ($n=2$) and Um ($n=7$). Box, first to last quartiles; whiskers, 1.5x interquartile range; center line, median; dots: individual data points. **c**, Ternary plots and bar plots depicting the mismatch directionality for selected rRNA modifications (Ψ , Am, Cm and Gm). Ψ rRNA modifications tend toward U-to-C mismatches, whereas Am, Cm and Gm modifications did not show specific mismatch directionality patterns. See also Supplementary Figs. 2 and 3.

current intensity values of 15-mer regions that contained the modified site showed that the reads clustered into two distinct populations: the first population mainly comprised unmodified reads from the snoRNA-depleted strain, whereas the second comprised reads



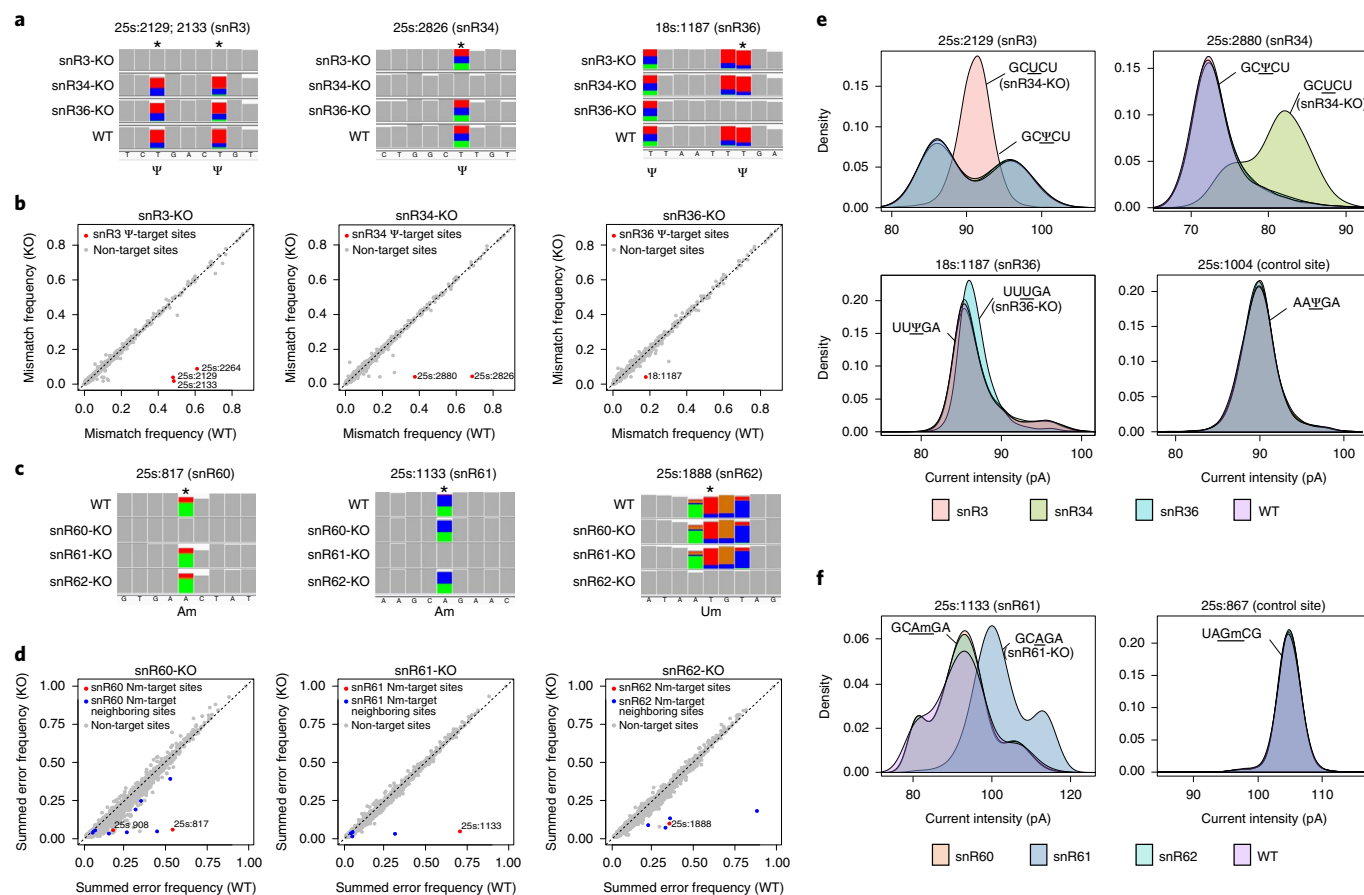


Fig. 3 | Pseudouridylation and 2'-O-methylations cause systematic base-calling 'errors' as well as altered current intensities, and their signature disappears upon depletion of snoRNAs guiding the modification. **a**, IGV snapshots of wild-type and three snoRNA-depleted strains depicting the site-specific loss of base-called errors at known Ψ target positions (indicated by asterisks). Nucleotides with mismatch frequencies greater than 0.1 are colored. **b**, Comparison of snoRNA knockout mismatch frequencies for each base, relative to wild-type, with snoRNA target sites indicated in red and non-target sites in gray. **c**, IGV snapshots of wild-type and three snoRNA knockout yeast strains depicting the site-specific loss of base-calling errors at known Nm target positions. Nucleotides with mismatch frequencies greater than 0.1 are colored. **d**, Comparison of snoRNA knockout summed error frequencies for each base, relative to wild-type, with snoRNA target sites indicated in red, neighboring sites in blue and non-target sites in gray. **e, f**, Distributions of per-read current intensity at known Ψ -modified (**e**) and 2'-O-methylated (**f**) sites. Current intensities at Ψ and 2'-O-methylated positions were altered upon deletion of specific snoRNAs relative to wild-type, whereas no shift was observed in control sites. WT, wild-type; KO, knockout.

from the three other strains, which are mostly modified (Fig. 4c, right panels, and Supplementary Fig. 4e).

To our surprise, we observed that Nanopolish software did not resquiggle the reads evenly across sites. For example, it failed at resquigging the majority of reads in the region surrounding 25s:Ψ2264 (Supplementary Fig. 4d). Thus, we examined whether Tombo, which uses global resquigging instead of local resquigging, might overcome this limitation. We found that Tombo resquigging led to a global increase in the proportion of resquigged reads (Supplementary Fig. 5a). Moreover, Tombo showed a uniform proportion of resquigged reads along the same transcript, whereas Nanopolish showed a variable proportion of resquigged reads depending on the site. Notably, Tombo was equally effective at resquigging both modified and unmodified reads, whereas Nanopolish preferentially resquigged unmodified reads relative to modified ones, biasing the unmodified:modified proportion up to 7:1 (Supplementary Fig. 5b). This uneven resquigging from Nanopolish implies that using Nanopolish for predicting RNA modification levels at individual sites might cause a dramatic bias in the predicted stoichiometry of individual sites. Thus, based on these results, we decided to adopt Tombo resquigging instead of Nanopolish resquigging for the prediction of RNA modification

stoichiometries from individual RNA reads in all our downstream analyses.

Stoichiometry prediction using signal intensity, dwell time and trace. Ψ and Nm modifications can lead to alterations in the current intensity profiles at the modified region (for example, 25s:Ψ2880; Fig. 4b,c). However, in other sites, such as 18s:Ψ1187, current intensity alone was insufficient to bin the reads into two separate clusters (Supplementary Fig. 4d,e), suggesting that, in addition to current intensity, other features might be needed to distinguish modified from unmodified reads.

Previous works predicting DNA modifications from individual nanopore reads typically relied on features such as signal intensity or dwell time to distinguish modified and unmodified read populations^{47–50}. In our study, in addition to these two features, we explored whether the use of 'trace' would improve our ability to predict RNA modification stoichiometry. Trace (also termed 'base probability'), which is reported by the Guppy base-calling algorithm, represents the probability that a given signal intensity chunk might be originating from each of the four canonical bases (A, C, G and T/U). To this end, we first examined how the presence of Ψ and Nm modifications altered each of the features (signal intensity, dwell time and trace) in

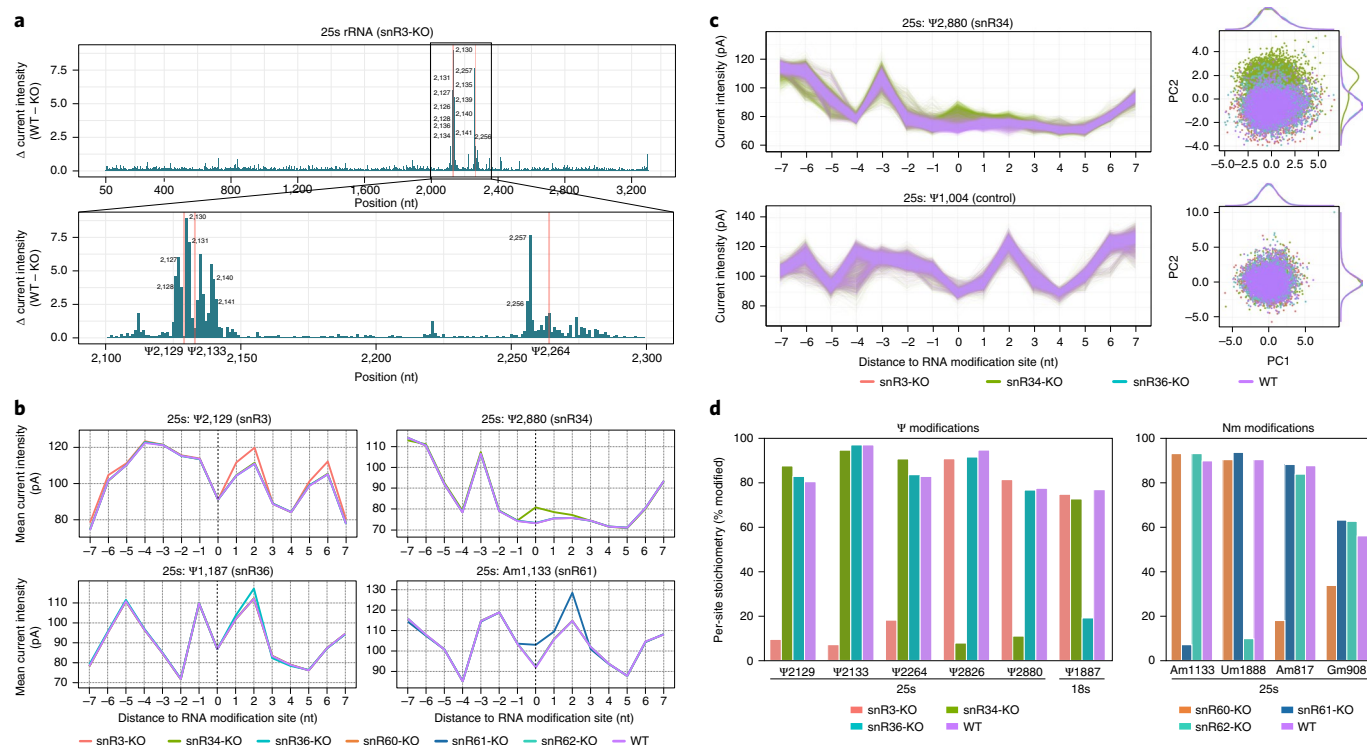


Fig. 4 | Loss of specific Ψ rRNA modifications causes deviations in current intensity in regions surrounding the Ψ sites. a, Current intensity changes along the 25s rRNA molecule upon snR3 depletion, relative to the wild-type strain. In the lower panel, a zoomed subset focusing on the two regions with the most significant current intensity deviations is shown—the first one comprising the 25s:Ψ2129 and 25s:Ψ2133 sites and the second one comprising the 25s:Ψ2264 site. **b**, Comparison of current intensities in the 15-mer regions surrounding Ψ and 2'-O-methyl knockout sites, for each of the four strains. The dotted vertical line indicates the modified position. See also Supplementary Fig. 4 for current intensity changes in other knockout strains and sites. **c**, Per-read current intensity analysis centered at the 25s:Ψ2880 site targeted by snR34 (upper panel) and a control site, 25s:Ψ2880, which is not targeted by any of the knockouts (lower panel). For each site, PCA was performed using 15-mer current intensity values, and the corresponding scatter plot of the two first principal components (PC1 and PC2) is shown on the right, using as input the same read populations as in the left panels. Each dot corresponds to a different read and is colored according to the strain. **d**, Predicted stoichiometry of Ψ- and Nm-modified sites using a KNN algorithm trained to classify the reads into two classes: modified or unmodified. The features used to predict modification status of every read from which stoichiometry was calculated were signal intensity (positions -1, 0 and +1) and trace (positions -1, 0 and +1). See also Supplementary Figs. 4 and 5. WT, wild-type; KO, knockout.

Ψ- and Nm-modified sites, both at snoRNA-targeted positions and control sites (Supplementary Fig. 6). We found that, in addition to signal intensity, base probability (trace) was significantly different in all examined sites. Moreover, in some sites, such as 25s:Ψ2264, trace was the most altered feature from those examined. By contrast, dwell time was not consistently different in snoRNA-targeted sites relative to wild-type (for example, 25s:Ψ2264, 25s:Ψ2826 and 18s:Ψ1187) (Supplementary Fig. 6).

We then proceeded to systematically benchmark the use of distinct features for RNA modification stoichiometry. To this end, we built nanoRMS, a software that extracts signal intensity, trace and dwell time from individual reads and then predicts RNA modification stoichiometry by using distinct feature combinations as well as various machine learning algorithms. First, we generated different mixes of modified (wild-type) and unmodified (knockout) reads to simulate varying read stoichiometry (0%, 20%, 40%, 60%, 80% and 100%) for each of the Ψ and Nm positions for which knockouts were available (Supplementary Table 3). Then, using distinct combinations of the three features (signal intensity, trace and dwell time) for each individual site, we examined how different supervised and unsupervised algorithms would predict the stoichiometry of each of the sites (Supplementary Fig. 5c). Our results show that the combination of signal intensity and trace outperformed all the other feature combinations for predicting both Ψ and Nm modification

stoichiometry and that the supervised *k*-nearest neighbor (KNN) was the best-performing algorithm. The *k*-means clustering algorithm (KMEANS) was the best-performing algorithm among the unsupervised clustering methods tested, although its performance in predicting Ψ modification stoichiometry was slightly better than in the case of Nm modification stoichiometry predictions. Overall, we found that nanoRMS can accurately predict Ψ and Nm RNA modification stoichiometry from individual RNA reads (Fig. 4d), with predicted stoichiometry values that are similar to those that were previously reported by mass spectrometry⁵¹ (Supplementary Table 4).

De novo prediction reveals a PUS4-dependent mitochondrial Ψ rRNA modification. The identification of RNA modification-specific signatures allows us to perform de novo prediction of Ψ RNA modifications transcriptome-wide using direct RNA sequencing. In this regard, *S. cerevisiae* mitochondrial rRNAs remain much less characterized than cytosolic rRNAs, with only three modified sites identified so far in *S. cerevisiae* large subunit rRNAs (21s)⁵² and none in small subunit rRNAs (15s). Thus, we hypothesized that direct RNA might reveal previously uncharacterized Ψ-modified sites in mitochondrial rRNAs. To this end, we first determined the 'error'-based thresholds (mismatch frequency and C mismatch frequency) that would distinguish unmodified uridines from pseudouridines in cytosolic rRNAs (Fig. 5a). We then applied this

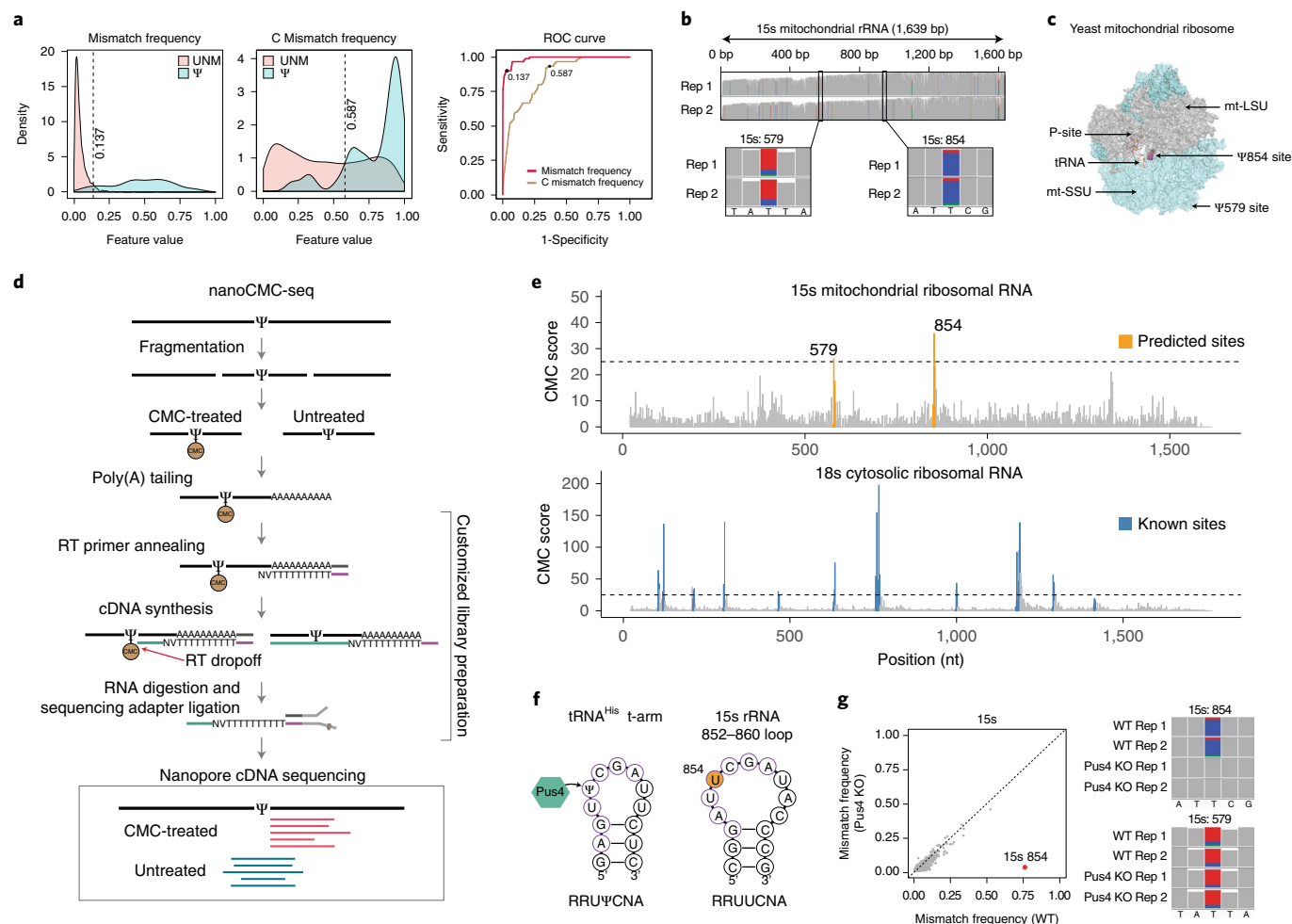


Fig. 5 | De novo prediction of Ψ modifications reveals a novel Pus4-dependent mitochondrial rRNA modification. **a**, Density distributions of mismatch and C mismatch frequency in unmodified uridine (red) and pseudouridine (cyan) positions. The dashed lines represent the optimal cutpoints between two groups determined by maximizing the Youden index. In the right panel, the ROC curve illustrates the sensitivity and specificity at these two cutpoints. **b**, IGV coverage tracks of the 15s mitochondrial rRNA, including a zoomed version showing the tracks centered at the 15s:854 and 15s:579 sites, in two biological replicates. Nucleotides with mismatch frequencies greater than 0.15 are colored. **c**, Location of the putative Ψ 854-modified site in the yeast mitochondrial ribosome. The PDB structure shown corresponds to 5MRC. **d**, Validation of putative Ψ sites with nanoCMC-Seq, which combines CMC treatment with nanopore cDNA sequencing to capture RT-drops that occur at Ψ -modified sites upon CMC probing. RT-drops are defined by counting the number of reads ending (3') at a given position. **e**, Predicted Ψ sites U854 and U579 (orange) in the 15s rRNA are validated using nanoCMC-seq (upper panel). Dashed lines indicate the CMC score threshold used for determining the positive sites (upper panel). As a control, we analyzed the nanoCMC-seq results in other rRNAs (lower panel), finding that all positions with a significant CMC score (>25) correspond to known Ψ rRNA modification sites (blue). See also Supplementary Fig. 7a for CMC scores in additional rRNA transcripts. **f**, The candidate Ψ 854 site is located at the 852–860 loop of the 15s rRNA, which resembles the t-arm of the tRNAs that is modified by Pus4. The binding motif of Pus4 (RRUUCNA) matches the motif surrounding the 854U site⁴. **g**, Scatter plot of mismatch frequencies in wild-type and Pus4 knockout cells, showing that the only significant position affected by the knockout of Pus4 is 15s:U854 (left panel). IGV coverage tracks showing that Pus4 knockout leads to depletion of the mismatch signature in the 15s:854 position (right panel) but not at the 15s:579 position. LSU, large subunit; mt, mitochondrial; PDB, Protein Data Bank; ROC, receiver operating characteristic; SSU, small subunit; WT, wild-type.

filter to predict Ψ modifications on 15s and 21s rRNAs, identifying two novel candidate Ψ sites (15s:854 and 15s:579) that displayed high modification frequency as well as U-to-C mismatch signature (Fig. 5b,c).

To further confirm that the two predicted 15s rRNA sites are pseudouridylated, we developed nanoCMC-seq, a protocol that identifies Ψ modifications by coupling CMC probing with nanopore cDNA sequencing. This method allows capturing reverse-transcription (RT) drop-off information by sequencing only the first-strand cDNA molecules of CMC-probed RNAs using a customized direct cDNA sequencing protocol (Fig. 5d and Methods). NanoCMC-seq captured known sites in cytoplasmic

rRNA with a very high signal-to-noise ratio, as well as confirming the existence of Ψ in positions 854 and 579 of 15s rRNA, validating our de novo predictions using direct RNA sequencing (Fig. 5e and Supplementary Fig. 7a).

We observed that 15s: Ψ 854 was embedded in a similar sequence context and structure as the t-arm of tRNAs, which contains a pseudouridylated (Ψ 55) position placed by Pus4 (Fig. 5f). Given the resemblance between these two sequences and structures, we hypothesized that Pus4 might be responsible for this modification. To validate our hypothesis, we sequenced total RNA from a Pus4-deficient *S. cerevisiae* strain, finding that the 15s:854 position loses its mismatch signature upon deleting the Pus4 gene,

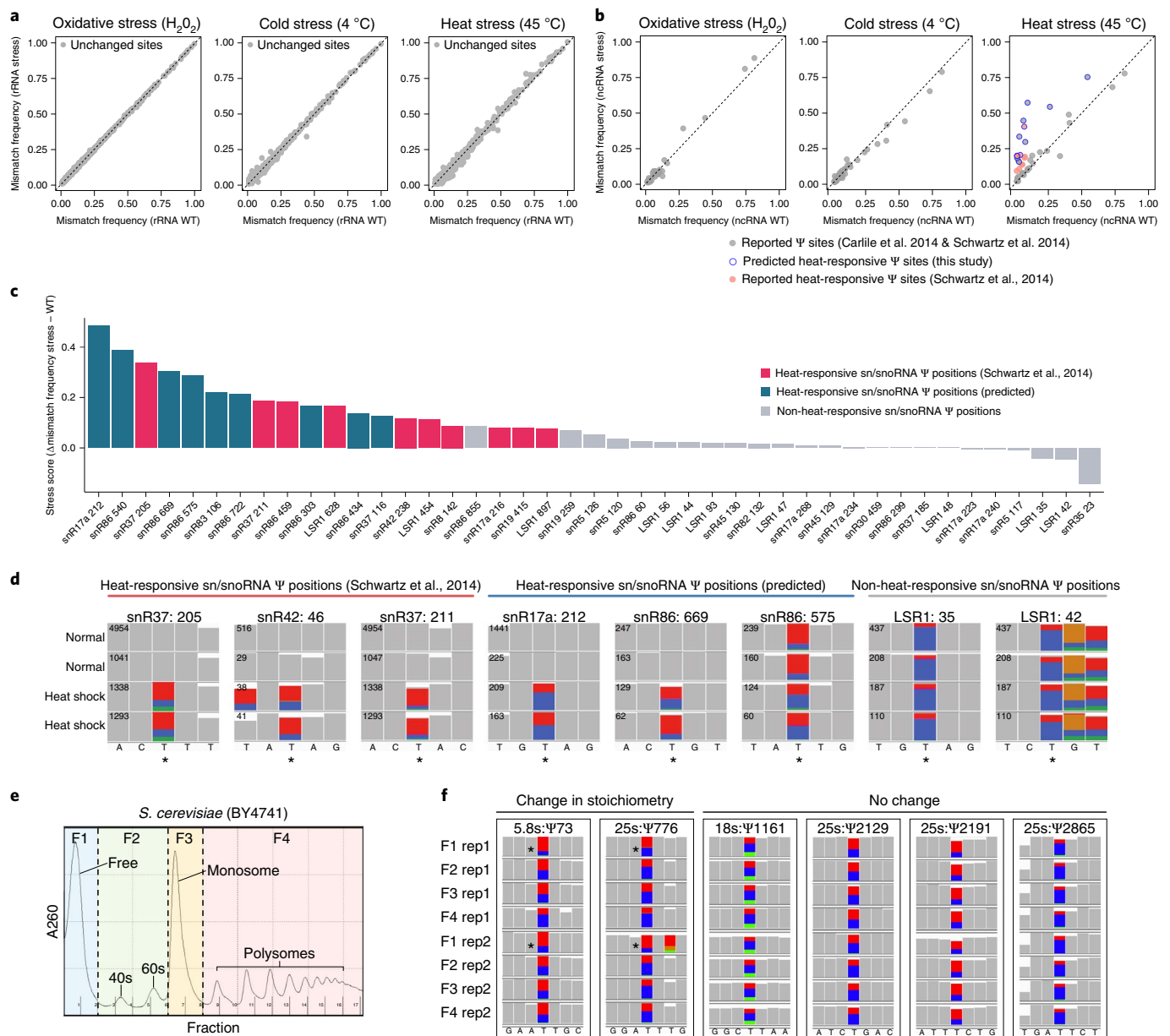


Fig. 6 | Comparative analysis of yeast rRNA and snRNA Ψ modifications upon distinct environmental stresses identifies known and previously unknown heat-sensitive snRNA and snoRNA Ψ modifications. **a**, Comparison of mismatch frequencies for all rRNA bases from untreated or yeast exposed to oxidative stress (H_2O_2 , left panel), cold stress ($4^\circ C$, middle panel) or heat stress ($45^\circ C$, right panel). Each dot represents a uridine base. All rRNA bases from cytosolic rRNAs were included in the analyses. **b**, Comparison of mismatch frequencies in untreated versus stressed-exposed yeast cells (oxidative, cold or heat) in previously reported ncRNA Ψ sites³⁴. **c**, Stress scores in sn/snoRNA Ψ sites calculated by Δ mismatch frequency between heat shock and wild-type. **d**, IGV snapshots of normal condition (rep1 and rep2) and heat shock condition (rep1 and rep2) yeast cells zoomed into the known sn/snoRNA Ψ positions (indicated by an asterisk). Nucleotides with mismatch frequencies greater than 0.1 are colored. Coverage for each position/condition is given on the top left of each row. **e**, Profiles of ribosomal fractions isolated from yeast grown under normal conditions, using sucrose gradient fractionation, including free rRNAs that are not assembled into ribosomal subunits (F1), rRNAs from 40s and 60s subunits (F2), rRNAs extracted from monosomal fractions (F3) and polysome fractions (F4). **f**, IGV snapshots of the two Ψ sites that change stoichiometry between translational fractions and four representative Ψ sites that show no significant change. Nucleotides with mismatch frequencies greater than 0.1 are colored. See also Supplementary Figs. 7 and 8. WT, wild-type.

confirming that this site is not only pseudouridylated but also that it is Pus4 dependent (Fig. 5g and Supplementary Fig. 7b). Additionally, we observed that previously reported Pus4 target sites (TEF1:239 and TEF2:239)^{3–5} completely lost their mismatch signature in Pus4 knockout cells (Supplementary Fig. 7b,c), confirming that our method is able to capture previously reported Pus4-dependent Ψ sites in addition to previously unknown ones.

rRNA modification profiles do not vary upon oxidative or thermal stress. rRNAs are extensively modified as part of their normal maturation, and their modification landscape is relatively well defined for a series of organisms^{39,53–57}. Despite the central role that rRNA molecules play in protein translation, recent evidence has shown that rRNA modifications are, in fact, dynamically regulated^{58,59} and that their alterations can lead to disease

states^{41,42,60–66}. Moreover, the stoichiometry of some pseudouridylated and 2'-O-methylated rRNA sites is cell type dependent, suggesting that rRNA modifications might be an important source of ribosomal heterogeneity^{43,51,54,67–69}. However, a systematic and comprehensive analysis of which environmental cues might lead to changes in rRNA modification stoichiometries, which RNA modifications might be subject to this tuning and to which extent this occurs is largely missing.

To assess whether rRNA modification profiles change in response to environmental stimuli, we treated *S. cerevisiae* cells with diverse environmental cues (oxidative, cold and heat stress) and sequenced their RNA using direct RNA sequencing. First, we confirmed that the rRNA modification profiles from independent biological replicates were highly reproducible (Pearson's $r^2 = 0.976–0.996$; Supplementary Fig. 8). Then, we examined whether exposure to stress would lead to significant changes in base-calling 'errors' in rRNA molecules, finding no significant differences in rRNA modification profiles between normal and stress conditions (Fig. 6a). In contrast, we recapitulated previously reported changes in snRNA Ψ modifications upon exposure to environmental cues⁴ (Fig. 6b and Supplementary Fig. 7d) as well as identified eight additional Ψ modification sites in snRNAs and snoRNAs whose stoichiometry varies upon heat exposure, which had not been previously described (Fig. 6, Supplementary Fig. 7e and Supplementary Table 5)^{3,4,38,70}. Overall, our approach confirmed previous reports and predicted novel Ψ sites in ncRNAs whose modification levels vary upon heat shock exposure (Fig. 6b–d and Supplementary Fig. 7d,e) but did not identify any rRNA modified site to be varying in its stoichiometry upon any of the tested stress conditions.

rRNA modification profiles do not vary across translational repertoires. Next, we questioned whether pseudouridylation changes in distinct translational repertoires might be more nuanced, in that Ψ levels might differ between rRNAs present in different translational fractions along a polysome gradient, which would not be detected when examining rRNAs as a whole. To test this, we sequenced both total (input) and polysomal rRNAs from untreated and H_2O_2 -treated yeast cells (Supplementary Fig. 7f). However, we observed no significant changes in Ψ rRNA modification profiles when comparing rRNAs from actively translating ribosomes in untreated versus H_2O_2 -treated cells (Supplementary Fig. 7g).

In an attempt to further dissect the different translational repertoires into a higher number of rRNA pools, we sequenced (1) rRNAs from unassembled free rRNA fractions (F1), (2) rRNAs from 40s and 60s subunits (F2), (3) rRNAs from monosomal fractions (F3) and (4) rRNAs from polysomal fractions (F4) (Fig. 6e). Although

two positions showed slightly decreased levels of Ψ (5.8s: Ψ 73 and 25s: Ψ 776) in the free rRNA fraction (F1) compared to assembled ribosomes and/or subunits, no significant changes were observed across the other translational fractions (Fig. 6f and Supplementary Fig. 7h). Globally, these results indicate that differential rRNA modifications are likely not the mechanism employed by yeast cells to adapt to environmental stress conditions, in agreement with previous observations³.

De novo prediction of Ψ modifications in mRNAs. rRNAs are modified at very high stoichiometries^{51,54}. By contrast, other molecules, such as mRNAs, are modified at lower stoichiometries, making the detection of their RNA modifications a much more challenging task²¹. To ascertain whether our methodology would be applicable to lowly modified RNA sites, such as those present in mRNAs, we first assessed the performance of nanoRMS in RNA molecules that contained Ψ RNA modifications at low RNA modification stoichiometries (0%, 3%, 7% and 20%) (Fig. 7a and Methods). The relative incorporation of Ψ RNA modifications was validated using mass spectrometry. We then examined the quantitative performance of nanoRMS under low stoichiometry conditions using both KNN and k -means, finding that the combination of signal intensity and trace features yielded the most accurate results in terms of stoichiometry prediction (Fig. 7b), in agreement with our previous results (Supplementary Fig. 5c).

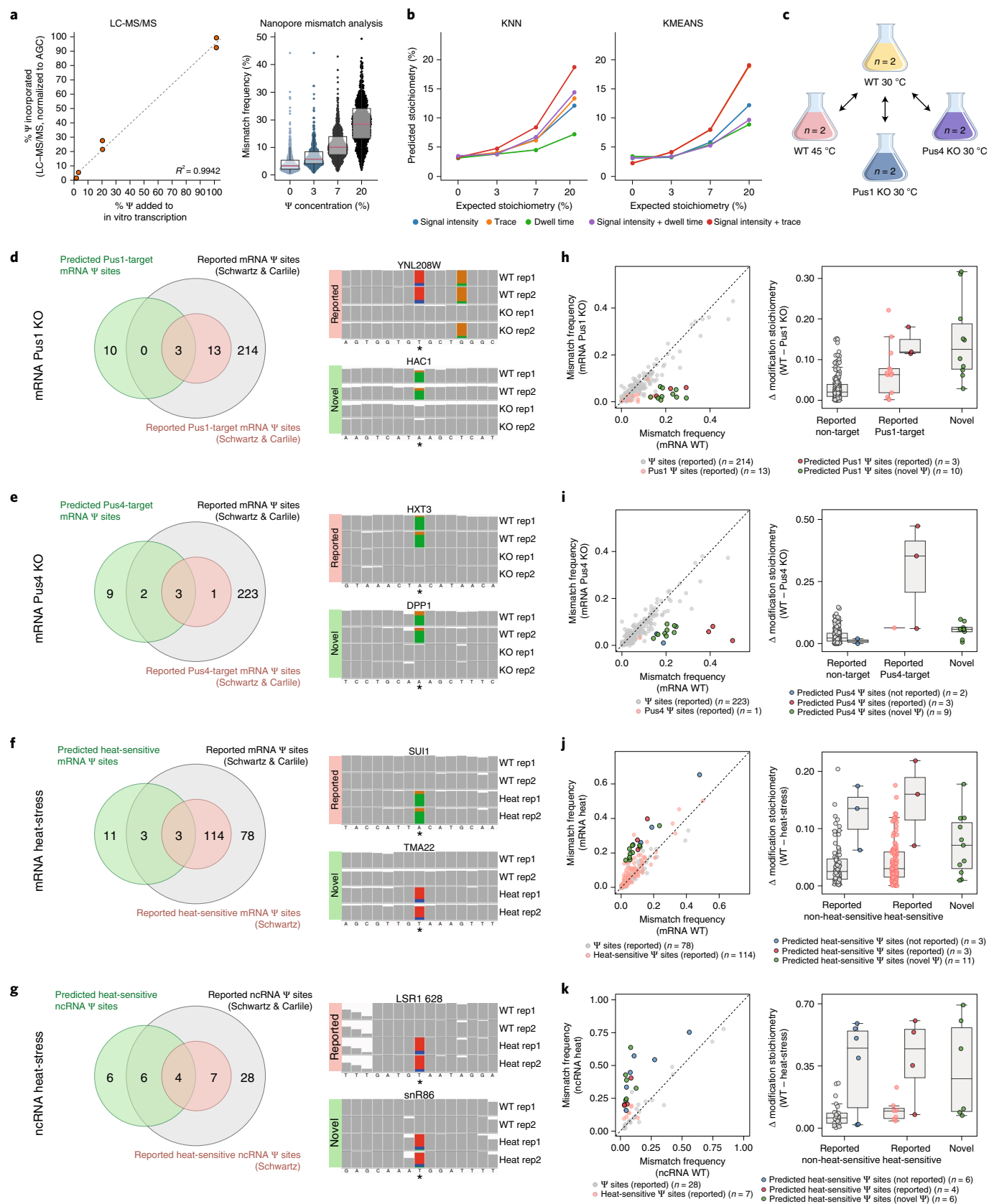
Next, we sequenced poly(A)⁺-selected RNA from *S. cerevisiae* wild-type, Pus1 knockout, Pus4 knockout and heat stress-exposed strains using direct RNA sequencing in biological duplicates. Considering that mRNA sites are lowly modified, we restricted our de novo identification of mRNA Ψ sites to those whose base-calling 'error' features significantly changed between pairwise conditions (Fig. 7c and Methods), met the pseudouridine 'error' signature and had a minimum coverage of 30 reads in both conditions and biological replicates (Supplementary Table 6 and Methods). Through this approach, we predicted 13 Pus1-dependent Ψ mRNA modifications, 14 Pus4-dependent Ψ mRNA modifications, 17 heat stress-dependent Ψ mRNA modifications and 16 heat stress-dependent Ψ ncRNA modifications, respectively (Fig. 7d–g, left panels, and Supplementary Tables 7–10), some of which were not previously reported to be Ψ modified.

NanoRMS recovered 11% of previously reported Pus1-dependent Ψ sites as well as 75% of Pus4-dependent Ψ sites, in addition to predicting 10 not previously reported Pus1-dependent mRNA Ψ -modified sites and 11 not previously reported Pus4-dependent mRNA Ψ -modified sites (Supplementary Tables 7 and 8). These novel predicted Ψ mRNA sites displayed similar mismatch

Fig. 7 | Quantitative prediction of pseudouridine stoichiometry transcriptome-wide and systematic benchmarking of nanoRMS using RNA molecules with diverse modification stoichiometries. **a**, LC-MS/MS validation of pseudouridine incorporation at different proportions (0%, 3%, 20% and 100%) in the in vitro transcribed products, relative to the expected incorporation (% Ψ TP relative to UTP) (left panel). Dot plot illustrates the mismatch frequency distribution of the uridine positions in the in vitro transcribed products incorporated with different concentrations of Ψ (right panel). Each dot represents one uridine position. **b**, Stoichiometry predictions of the Ψ incorporated in vitro transcription products using two different algorithms (KNN and k -means) with different current information (middle-right and right panels). **c**, Conditions and strains used to predict Ψ mRNA modifications transcriptome-wide. **d–k**, Transcriptome-wide Ψ RNA modification predictions and predicted stoichiometries in mRNAs and ncRNAs, for Pus1-dependent mRNA Ψ sites (**d**, **h**), Pus4-dependent mRNA Ψ sites (**e**, **i**), heat stress-dependent mRNA Ψ sites (**f**, **j**) and heat stress-dependent ncRNA Ψ sites (**g**, **k**). **d–g**, Venn diagrams depict the overlap between Ψ sites predicted by our analysis and the previously reported pseudouridine sites. IGV snapshots of reported and not previously reported predicted sites illustrate the absence of the mismatch signature in the Pus1 (**d**) or Pus4 (**e**) knockout samples as well as under normal conditions, relative to heat stress conditions in mRNA (**f**) and ncRNA (**g**). The reported or predicted Ψ site is indicated by an asterisk. Nucleotides with mismatch frequencies greater than 0.15 are colored. We should note that IGV snapshots that show a reference 'A' with mismatch signature to G are genes that are in the minus strand (and, thus, are, in reality, positions showing U-to-C mismatch signatures). **h–k**, Quantitative analysis of previously reported and de novo predicted Ψ sites in mRNAs and ncRNAs. In the left panels, comparative scatter plots of mismatch frequency illustrate differentially modified sites of reported and de novo predicted Ψ sites. In the right panels, stoichiometry prediction differences between wild-type and knockout strains (**h**, **i**) or between normal and heat stress conditions (**j**, **k**) are shown as box plots. Box, first to last quartiles; whiskers, 1.5 $\times$ interquartile range; center line, median; points, individual Ψ sites. See also Supplementary Fig. 9. LC-MS/MS, liquid chromatography with tandem mass spectrometry; WT, wild-type; KO, knockout.

signatures to those observed in previously reported Ψ sites (Fig. 7d,e, right panels) and were highly replicable across biological replicates, and their signature disappeared in Pus1 or Pus4 knockout strains. Similarly, nanoRMS was able to capture previously

reported heat-responsive Ψ sites present in mRNAs and ncRNAs, which resulted in predicting 17 heat-responsive Ψ mRNA sites, of which six were previously reported Ψ sites (Fig. 7f and Supplementary Table 9), as well as 16 heat-responsive Ψ ncRNA



sites, of which 10 were previously reported Ψ sites (Fig. 7g and Supplementary Table 10).

Surprised by the relatively poor overlap between our predictions and previously reported Pus1 mRNA Ψ -modified sites (3 out of 16 sites), as well as between predicted and previously reported heat stress-dependent sites (7 out of 128 sites), we inspected the individual per-read features at previously reported Pus1- and heat stress-dependent sites (Supplementary Fig. 9a,b). Indeed, the Ψ sites that nanoRMS did not report as Pus1- or heat stress-dependent were not significantly different for any of the features examined (current intensity, dwell time or trace). Thus, we wondered whether some of these sites might have been misassigned as Pus1- or heat stress-dependent by previous works. A closer examination of the overlap between Ψ sites predicted by the two previously published studies using CMC probing coupled to Illumina sequencing^{3,4}, which we used to define the set of ‘previously reported Pus1-, Pus4- and heat stress-dependent Ψ sites’, showed that the overlap was, in fact, very poor (Supplementary Fig. 9c), both when examining the set of predicted mRNA and ncRNA Ψ sites (7% and 17%, respectively) and when examining the sets of predicted Pus1- and Pus4-dependent mRNA and ncRNA Ψ sites (6% and 50%, respectively). Altogether, our approach detected 100% of Pus1- and Pus4-dependent sites that were identified by both studies but very few of those that were identified by only one of the studies. Thus, we conclude that the poor overlap between our results and previously reported Ψ sites is, in fact, a direct consequence of the poor overlap between the set of predicted Pus1-, Pus4- and heat stress-dependent mRNA and ncRNA Ψ sites by the two previous studies (Supplementary Fig. 9c). The relatively low overlap of predicted sites between studies may reflect a combination of false positives, false negatives and possibly also differences in growth conditions.

Finally, we applied nanoRMS to predict the modification stoichiometry of all Ψ sites predicted in mRNAs and ncRNAs. Reads were classified based on the per-read signal intensity and trace features from positions -1 , 0 and $+1$ using the k -means unsupervised clustering algorithm (Fig. 7h–k). As expected, per-read stoichiometry predictions were low in non-targeted Ψ sites. By contrast, predicted Ψ Pus1/Pus4/heat stress-dependent sites (which included all Ψ sites) typically showed significant RNA modification stoichiometry changes, ranging from 5% to 50% change in their Ψ modification stoichiometries between the two conditions.

Altogether, we find that differential ‘error’ Ψ signatures are a useful approach to identify dynamic Ψ RNA modifications across two conditions, even at low stoichiometry sites, and that nanoRMS can be used to de novo predict and quantify the RNA modification stoichiometry dynamics, both in previously reported Ψ sites and in de novo predicted Ψ sites.

Discussion

RNA modifications regulate a wide range of biological processes^{71–73}. They can modulate the fate of RNA molecules by altering mRNA splicing^{74–76} or mRNA decay^{77,78} as well as affect major cell and organism-level decisions, such as cellular differentiation^{79,80} and sex determination^{18,81,82}. Although the biological relevance of RNA modifications is beyond question, a major difficulty in studying them has been the need for tailored protocols to map each modification individually^{20,83}. In this context, direct RNA nanopore sequencing can overcome many of the limitations that NGS-based methods suffer from, as it can sequence full-length native RNA molecules, including their RNA modifications.

Direct RNA nanopore sequencing has been successfully applied in a wide variety of organisms^{29–31,84–87}. However, the detection of distinct RNA modification types in individual native RNA molecules is still an unsolved challenge. While both current intensity-based and ‘error’-based methods have proven useful strategies to detect RNA modifications, these have been mainly focused on the detection of

m^6A ^{29–31,33} and are typically unable to predict which RNA modification type they are in fact detecting (for example, m^6A , Ψ , Am or m^5C)^{28,50}. Moreover, current algorithms to study RNA modifications using direct RNA sequencing are not quantitative.

To overcome these limitations, in this study we first explored how distinct RNA modifications might affect direct RNA nanopore signals and base-calling ‘errors’. We found that different RNA modification types (for example, Ψ versus m^5C) produce distinct, yet characteristic, base-calling ‘error’ signatures, both in vitro (Fig. 1 and Supplementary Fig. 1f) and in vivo (Fig. 2). Consequently, base-calling errors can be used not only to predict whether a given site is modified but also to identify the underlying RNA modification type. Although base-calling signatures depend, to some extent, on the surrounding sequence context, we found that Ψ modifications lead to robust U-to-C mismatch signatures, which can be exploited for de novo prediction of Ψ modifications (Fig. 5). Through this approach, we identified two previously unreported Ψ modifications in yeast 15s mitochondrial rRNA (15s:579 and 15s: Ψ 854) as well as confirmed reported Ψ -modified sites in rRNAs, snRNAs and mRNAs (Figs. 3–7). Moreover, we revealed that Pus4, which was previously thought to modify only tRNAs and mRNAs, is the enzyme responsible for placing Ψ 854 in mitochondrial rRNA. These findings were further validated using nanoCMC-seq, a novel orthogonal method that can detect Ψ modifications with single-nucleotide resolution by coupling CMC probing to nanopore cDNA sequencing (Fig. 5d).

We found that Ψ modifications can be detected both in the form of base-calling ‘errors’ and altered current intensities (Figs. 3 and 4); however, we observed that the latter does not provide single-nucleotide resolution, with maximal current intensity shifts often seen a few nucleotides away from the real modified site. Thus, current intensity-based methods alone might suffer from imprecisions in the assignment of the RNA-modified site. Here we propose that the combination of both approaches is the optimal design to obtain stoichiometric information of Ψ -modified sites with single-nucleotide resolution. Specifically, we show that, once the site has been located using base-calling error features, per-read features (current intensity and trace) from the regions surrounding the Ψ - or Nm-modified site are sufficient to robustly bin the reads into two separate clusters (modified and unmodified) and provide good estimates of Ψ and Nm modification stoichiometries (Figs. 4d and 7b).

One surprising feature of base-calling ‘errors’ is that fully modified sites do not always lead to the same mismatch frequencies, suggesting that mismatch frequencies alone cannot be used, per se, as an estimation of the stoichiometry of the site (Fig. 2b). While, within the same sequence context, higher mismatch frequencies correspond to higher modification levels, this same rule cannot be used to compare across distinct RNA modified sites. We speculate that the differences observed in mismatch frequency across different sites might be, in fact, a consequence of the distinct deviations in current intensity of the modified k -mer relative to unmodified counterparts (Supplementary Fig. 9d).

Finally, we should note that, while nanoRMS allows predicting and studying the dynamics of diverse RNA modifications in a quantitative manner, there are caveats and limitations, leaving ample room for future improvements. First, not all RNA modifications lead to strong alterations in the base-calling features and/or current intensity patterns, such as 2′-O-methylcytosine (Cm), which is poorly detected in direct RNA sequencing datasets compared to other RNA modifications (Fig. 2c). Second, the detection of RNA modifications is partly dependent on the sequence context; for example, we were unable to detect 25s:Gm908 (Supplementary Fig. 3). Similarly, some Ψ -modified sites, such as 18s: Ψ 1187, cause weaker alterations in base-calling features and current intensity shifts than other Ψ -modified positions (Figs. 3 and 4), although this limitation can be alleviated by the incorporation of additional

features into the model (Supplementary Fig. 5c). Third, not all RNA modifications lead to base-calling errors with single-nucleotide resolution, as with pseudouridine. For example, 2'-O-methylations often affect neighboring bases (Fig. 3c and Supplementary Fig. 4a), making it challenging to de novo predict modified sites without any prior information. Fourth, stoichiometry prediction is heavily affected by the choice of resquigging algorithms (Supplementary Figs. 5 and 10). For example, we were unable to predict stoichiometry in 25s:Ψ2264 when using resquigging due to the low number of reads that the Nanopolish algorithm was able to resquiggle (Supplementary Fig. 4e); however, this limitation could be overcome when using Tombo resquigging, leading to stoichiometry predictions similar to those observed using mass spectrometry (Fig. 4d). Finally, we should note that, although nanoRMS was successful at detecting RNA modification stoichiometry changes as low as 5–10% (Fig. 7), the detection of RNA modification changes in low-modification stoichiometry sites was possible only when using pairwise comparisons.

Despite these challenges and limitations, our work provides a framework for the systematic and comprehensive analysis of the epitranscriptome with single-molecule resolution, showing that direct RNA sequencing can be employed to estimate Ψ and Nm modification stoichiometry as well as to de novo predict Ψ RNA modifications transcriptome-wide in rRNAs, ncRNAs and mRNAs. Future work will be needed to functionally dissect the biological roles and dynamics of RNA modifications, to better comprehend how and when the epitranscriptome is tuned to regulate diverse cellular functions.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41587-021-00915-6>.

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Methods

Yeast culturing. *S. cerevisiae* (strain BY4741) was grown at 30 °C in standard YPD medium (1% yeast extract, 2% Bacto Peptone and 2% dextrose). The deletion strains snR3Δ, snR34Δ and snR36Δ (Ψ-deficient in specific rRNA positions), as well as snR60Δ, snR61Δ and snR62Δ (Nm-deficient in specific rRNA positions) were generated on the background of the BY4741 strain by replacing the genomic snoRNA sequence with a kanMX4 cassette, as detailed in Parker et al.⁸⁸ Cells were then quickly transferred into 50-ml pre-chilled Falcon tubes and centrifuged for 5 min at 3,000g in a 4 °C pre-chilled centrifuge. Supernatant was discarded, and cells were flash-frozen. For thermal stress, *S. cerevisiae* BY4741 cultures were grown in 4 ml of YPD overnight at 30 °C. The next day, cultures were diluted to 0.0001 OD₆₀₀ in 200 ml of YPD and grown overnight at 30 °C shaking (250 r.p.m.). When the cultures reached an OD₆₀₀ of 0.4–0.5, the cultures were divided into 3 × 50-ml subcultures, which were then incubated at 30 °C (control), 45 °C (heat shock) or 4 °C (cold shock) for 1 h. Cells were collected by pelleting and snap-freezing. For the analysis of rRNA modifications across polysomal fractions, yeast BY4741 starter cultures were grown in 6 ml of YPD medium at 30 °C with shaking (250 r.p.m.) overnight. Next, 100 ml of fresh YPD medium was inoculated with 10 μl of the stationary culture in a 250-ml Erlenmeyer flask in biological duplicates. Cells were incubated at 30 °C with shaking (250 r.p.m.) until the cultures reached mid-exponential growth phase (OD₆₀₀ ~0.4–0.6). Yeast cells were then treated with 1 mM H₂O₂ or left without treatment (control) for 30 min. Then, 1 ml of cycloheximide stock solution (10 mg ml⁻¹) was added to each culture. Pus4 and Pus1 knockout strains (BY4741 MATa pus4::KAN, BY4741 MATa pus1::KAN) and the parental strain were obtained from the Yeast Knockout Collection (Dharmacon) and grown under standard conditions in YPD (1% (wt/vol) yeast extract, 2% (wt/vol) peptone supplemented with 2% glucose) at 30 °C unless stated otherwise.

Total RNA extraction from yeast cultures. *S. cerevisiae* BY4741 cells (strains: snR3Δ, snR34Δ, snR36Δ, snR60Δ, snR61Δ, snR62Δ and wild-type) were harvested via centrifugation at 3,000 r.p.m. for 1 min, followed by two washes with water. RNA was purified from pelleted cells using a MasterPure Yeast RNA extraction kit (Lucigen, MPY03100), according to the manufacturer's instructions. Total RNA was then treated with Turbo DNase (Thermo Fisher Scientific, no. AM2238) with a subsequent RNAClean XP bead cleanup before starting the library preparation. For stress conditions and the Pus4 knockout strain, flash-frozen pellets were resuspended in 700 μl of TRIzol with 350 μl of acid-washed and autoclaved glass beads (425–600 μm, Sigma-Aldrich, G8772). The cells were disrupted using a vortex on top speed for seven cycles of 15 s (the samples were chilled on ice for 30 s between cycles). Afterwards, the samples were incubated at room temperature for 5 min, and 200 μl of chloroform was added. After briefly vortexing the suspension, the samples were incubated for 5 min at room temperature. Then, the samples were centrifuged at 14,000g for 15 min at 4 °C, and the upper aqueous phase was transferred to a new tube. RNA was precipitated with 2× volume of molecular-grade absolute ethanol and 0.1× volume of sodium acetate. The samples were then incubated for 1 h at –20 °C and centrifuged at 14,000g for 15 min at 4 °C. The pellet was then washed with 70% ethanol and resuspended with nuclease-free water after air drying for 5 min on the benchtop. Purity of the total RNA was measured with a NanoDrop 2000 spectrophotometer. Total RNA was then treated with Turbo DNase (Thermo Fisher Scientific, no. AM2238) with a subsequent RNAClean XP bead cleanup.

mRNA extraction from yeast cultures. *S. cerevisiae* BY4741 (strains: BY4741 MATa pus4::KAN, BY4741 MATa pus1::KAN and BY4741 MATa) were cultured up to log-phase at 30 °C. The cultures were then divided into two flasks and cultivated at 30 °C or 45 °C for 1 h. The cells were harvested via centrifugation at 3,000 r.p.m. for 5 min and snap-frozen. Total RNA was purified from pelleted cells using a MasterPure Yeast RNA extraction kit, according to the manufacturer's instructions. Total RNA was then DNase-treated (Ambion, AM2239) at 37 °C for 20 min with a subsequent cleanup using RNeasy MinElute Cleanup Kit (Qiagen, 74204). Then, 70–100 μg of total RNA was subjected to double poly(A) selection using Dynabeads Oligo(dT)₂₅ (Invitrogen, 61002) and, finally, eluted in ice-cold 10 mM Tris pH 7.5.

Polysome gradient fractionation and rRNA extraction. Yeast pellets from 100-ml cultures were washed with 6 ml of ice-cold polysome extraction buffer (PEB), which contained 20 mM Tris-HCl pH 7.4, 100 mM KCl, 10 mM MgCl₂, 0.5 mM DTT, 0.1 mg ml⁻¹ of cycloheximide and 100 U ml⁻¹ of RNase inhibitors (RNaseOUT, Invitrogen, no. 18080051). Cells were centrifuged for 5 min at 3,000g at 4 °C. Washing was repeated by adding 6 ml of ice-cold PEB, followed by centrifugation. Cells were then resuspended in 700 μl of ice-cold PEB and transferred into pre-chilled 2-ml Eppendorf tubes containing 450 μl of pre-chilled RNase-free 425–600-μm-diameter glass beads (Sigma-Aldrich, G8772). Cells were lysed by vortexing at maximum speed for 5 min at 4 °C, followed by centrifugation, also at maximum speed, on a bench centrifuge for 5 min at 4 °C. Next, 10% of the supernatant was aliquoted into TRIzol for total RNA isolation and kept at –80 °C, which was later used as input. The remaining volume, corresponding approximately to 8 × 10⁸ cells, was subsequently loaded onto the sucrose gradient.

Linear sucrose gradients of 10–50% were prepared using the Gradient Station (BioComp). Briefly, SW41 centrifugation tubes (Beckman Coulter, Ultra-Clear™ 344059) were filled with Gradient Solution 1 (GS1), which consisted of 20 mM Tris-HCl pH 7.4, 100 mM KCl, 10 mM MgCl₂, 0.5 mM DTT, 0.1 mg ml⁻¹ of cycloheximide and 10% wt/vol RNase-free sucrose. Solutions GS1 and Gradient Solution 2 (GS2) were prepared with RNase/DNase-free UltraPure water and filtered with a 0.22-μm filter. The tube was then filled with 6.3 ml of GS2 layered at the bottom of the tube, which consisted of 20 mM Tris-HCl pH 7.4, 100 mM KCl, 10 mM MgCl₂, 0.5 mM DTT, 0.1 mg ml⁻¹ of cycloheximide and 50% wt/vol RNase-free sucrose. The linear gradient was formed using the tilted methodology, with the Gradient Station Maker (BioComp). Once the gradients were formed, 350 μl of each lysate was carefully loaded on top of the gradients, and tubes were balanced in pairs, placed into pre-chilled SW41Ti buckets and centrifuged at 4 °C for 150 min at 35,000 r.p.m. Gradients were then immediately fractionated using the Gradient Station, and 20 × 500-μl fractions were collected in 1.5-ml Eppendorf tubes while absorbance was monitored at 260 nm continuously. Fractions were combined in the following way: the free rRNA (F1, fractions 1 and 2), the unassembled subunits (F2, fractions 3–6), the lowly translating monosomes (F3, fractions 7–10) and the highly translating polysomes (F4, fractions 12–17). The pooled fractions were then concentrated using Amicon-Ultra 100K columns (Millipore) and washed two times with cold PEB. The final volume was brought down to 200 μl, and RNA was extracted using TRIzol reagent. Purity of the RNA was measured with a NanoDrop 2000 spectrophotometer.

In vitro transcription of modified and unmodified RNAs. The synthetic 'curlcake' sequences²⁹ used in this study are designed to include all possible 5-mers, while minimizing the secondary RNA structure, and consist in four in vitro transcribed constructs: (1) Curlcake 1, 2,244 bp; (2) Curlcake 2, 2,459 bp; (3) Curlcake 3, 2,595 bp and (4) Curlcake 4, 2,709 bp. The curlcake constructs were in vitro transcribed using the AmpliScribe T7-Flash Transcription Kit (Lucigen, ASF3507) with either unmodified rNTPs (UNM), N⁶-methyladenosine triphosphate (m⁶ATP), 5-methylcytosine triphosphate (m⁵CTP), 5-hydroxymethylcytosine triphosphate (hm⁵CTP) or pseudouridine triphosphate (ΨTP). All modified NTPs were purchased from TriLink. The sequences included in the short unmodified dataset (UNM-S), which included *Bacillus subtilis* guanine riboswitch, *B. subtilis* lysine riboswitch and *Tetrahymena* ribozyme, were also produced by in vitro transcription using the AmpliScribe T7-Flash Transcription Kit. All constructs were 5' capped using vaccinia capping enzyme (NEB, M2080S) and polyadenylated using *Escherichia coli* poly(A) polymerase (NEB, M0276S). Poly(A)-tailed RNAs were purified using RNAClean XP beads, and the addition of poly(A)-tail was confirmed using Agilent 4200 TapeStation. Concentration was determined using Qubit Fluorometric Quantitation. Purity of the in vitro transcription product was measured with a NanoDrop 2000 spectrophotometer.

Direct RNA library preparation and sequencing of in vitro transcribed constructs. The RNA libraries for direct RNA sequencing (SQK-RNA001) were prepared following the ONT Direct RNA Sequencing protocol version DRS_9026_v1_revP_15Dec2016, which corresponds to the flowcell FLO-MIN106. Briefly, 800 ng of poly(A)-tailed and capped RNA (200 ng per construct) was ligated to ONT RT Adaptor (RTA) using concentrated T4 DNA Ligase (NEB, M0202T) and was reverse transcribed using SuperScript III RT (Thermo Fisher Scientific, 18080044). The products were purified using 1.8× Agencourt RNAClean XP beads (Thermo Fisher Scientific, NC0068576), washing with 70% freshly prepared ethanol. RNA Adapter (RMX) was ligated onto the RNA:DNA hybrid, and the mix was purified using 1× Agencourt RNAClean XP beads, washing with wash buffer twice. The sample was then eluted in elution buffer and mixed with RNA running buffer before loading onto a primed R9.4.1 flowcell and run on a MinION sequencer with MinKNOW acquisition software version 1.15.1. The sequencing was performed on independent days and using a different flowcell for each sample (UNM, m⁶A, m⁵C, hm⁵C, Ψ and UNM-S).

Direct RNA library preparation and sequencing of yeast total RNAs and mRNAs. We performed direct RNA sequencing of two types of *S. cerevisiae* RNA inputs: (1) total RNA from *S. cerevisiae* and (2) poly(A)-selected RNA from *S. cerevisiae*. Yeast total RNAs were polyadenylated using *E. coli* poly(A) polymerase, following the commercial protocol, before starting the library prep. Yeast poly(A)-selected RNA was directly used as input to start the libraries because they already contain poly(A) tail. Four different direct RNA libraries were barcoded according to the protocol that we recently published⁸⁹. Custom RT adaptors (IDT) were annealed using the following conditions: custom oligo A and B (Supplementary Table 11) were mixed in annealing buffer (0.01 M Tris-Cl pH 7.5, 0.05 M NaCl) to the final concentration of 1.4 μM each in a total volume of 75 μl. The mixture was incubated at 94 °C for 5 min and slowly cooled down (–0.1 °C s⁻¹) to room temperature. RNA library for direct RNA sequencing (SQK-RNA002) was prepared following the ONT Direct RNA Sequencing protocol version DRS_9080_v2_revI_14Aug2019 with half reaction for each library until the RNA Adapter (RMX) ligation step. Per reaction (half), 250 ng total of yeast RNAs were ligated to pre-annealed custom RT adaptors (IDT)⁸⁹ using concentrated T4 DNA Ligase (NEB, M0202T) and was reverse transcribed using Maxima H

Minus RT (Thermo Fisher Scientific, EP0752), without the heat inactivation step. The products were purified using 1.8× Agencourt RNAClean XP beads (Thermo Fisher Scientific, NC0068576) and washed with 70% freshly prepared ethanol. Then, 50 ng of reverse-transcribed RNA from each reaction was pooled, and RMX adapter, composed of sequencing adapters with motor protein, was ligated onto the RNA:DNA hybrid, and the mix was purified using 1× Agencourt RNAClean XP beads, washing with wash buffer twice. The sample was then eluted in elution buffer and mixed with RNA running buffer before loading onto a primed R9.4.1 flowcell and run on a MinION sequencer with MinKNOW acquisition software version v.3.5.5.

NanoCMC-seq. CMC treatment was adapted from Schwartz et al.⁴ with minor changes. Briefly, 20 µg of total RNA was incubated in NEBNext Magnesium RNA Fragmentation Module at 94 °C for 1.5 min. The fragmented RNA was then incubated with either 0.3 M CMC dissolved in 100 µl of TEU buffer (50 mM Tris pH 8.5, 4 mM EDTA, 7 M urea) or 100 µl of TEU buffer (no CMC) for 20 min at 37 °C. Reaction was stopped with 100 µl of Buffer A (0.3 M NaOAc and 0.1 mM EDTA, pH 5.6), 700 µl of absolute ethanol and 1 µl of GlycoBlue (Thermo Fisher Scientific, AM9515). RNA in the stop solution was chilled on dry ice for 5 min and then centrifuged at maximum speed for 15 min at 4 °C. Supernatant was removed, and the pellet was washed with 70% ethanol. After air drying for a few minutes, the pellet was dissolved in 100 µl of Buffer A and mixed with 300 µl of absolute ethanol and 1 µl of GlycoBlue. After chilling on dry ice for 5 min, the solution was then centrifuged at maximum speed for 15 min at 4 °C. Supernatant was removed, and the pellet was washed with 70% ethanol. After washing, the pellet was air dried and resuspended in 40 µl of 50 mM sodium bicarbonate, pH 10.4, and incubated at 37 °C for 3 h. Then, RNA was mixed with 100 µl of Buffer A, 700 µl of ethanol and 1 µl of GlycoBlue overnight at −20 °C. The next day, the solution was centrifuged at maximum speed for 15 min at 4 °C, and the pellet was washed with 70% ethanol and dissolved in the appropriate amount of water after air drying. Unprobed and probed RNAs were treated with T4 Polynucleotide Kinase (NEB, M0201S), as described above, before proceeding with ONT Direct cDNA Sequencing.

Before starting the library preparation, 9 µl of 100 µM reverse transcription primer (Original ONT VNP: 5′-5Phos/ACTTGCCTGCTCGCTCTATCTTCTTTT TTTT TTTT TTTT TTTT TTTT VN-3′) and 9 µl of 100 µM complementary oligo (CompA: 5′-GAAGATAGAGCGACAGGCAAGTA-3′) were mixed with 1 µl of 0.2 M Tris pH 7.5 and 1 µl of 1 M NaCl. The mix was incubated at 94 °C for 1 min, and the temperature was ramped down to 25 °C (−0.1 °C s^{−1}) to pre-anneal the oligos. Then, 100 ng of poly(A)-tailed RNA was mixed with 1 µl of pre-annealed VNP + CompA, 1 µl of 10 mM dNTP mix, 4 µl of 5× RT Buffer, 1 µl of RNasin Ribonuclease Inhibitor (Promega, N2511), 1 µl of Maxima H Minus RT (Thermo Fisher Scientific, EP0742) and nuclease-free water up to 20 µl. The reverse transcription mix was incubated at 60 °C for 60 min and inactivated by heating at 85 °C for 5 min before moving onto ice. Next, RNase Cocktail (Thermo Fisher Scientific, AM2286) was added to the mix to digest the RNA, and the mix was incubated at 37 °C for 10 min. Then, the reaction was cleaned up using 1.2× AMPure XP Beads (Agencourt, A63881). To be able to ligate the sequencing adapters to the first strand, 1 µl of 100 µM CompA was again annealed to the 15-µl cDNA in a tube with 2.25 µl of 0.1 M Tris pH 7.5, 2.25 µl of 0.5 M NaCl and 2 µl of nuclease-free water. The mix was incubated at 94 °C for 1 min, and the temperature was ramped down to 25 °C (−0.1 °C s^{−1}) to anneal the complementary to the first-strand cDNA. Next, 22.5 µl of first-strand cDNA was mixed with 2.5 µl of Native Barcode (EXP-NBD104) and 25 µl of Blunt/TA Ligase Mix (NEB, M0367S) and incubated at room temperature for 10 min. The reaction was cleaned up using 1× AMPure XP beads, and the libraries were pooled into one tube that, finally, contained the 200-fmol library. The pooled library was then ligated to the sequencing adapter (AMII) using Quick T4 DNA Ligase (NEB, M2200S) at room temperature for 10 min, followed by 0.65× AMPure XP Bead cleanup using ABB Buffer for washing. The sample was then eluted in elution buffer and mixed with sequencing buffer and loading beads before loading onto a primed R9.4.1 flowcell and run on a MinION sequencer with MinKNOW acquisition software version 3.5.5.

Analysis of nanoCMC-seq. Reads were base-called with standalone Guppy version 3.6.1 with default parameters running in GPU, with the built-in demultiplexing tool of Guppy. Unclassified reads were then demultiplexed further using Porechop with --barcode_threshold 50 option (<https://github.com/rrwick/Porechop>). Then all the merged classified reads were mapped to cytosolic and mitochondrial rRNA sequences in *S. cerevisiae* using minimap2 default. A custom script was used to extract RT-drop signatures, and the RT-drop scores were plotted using ggplot2. All scripts used to process nanoCMC-seq data with RT-drop information are available in GitHub (https://github.com/novoalab/yeast_RNA_Mod). Notably, due to the 5′ end truncation of the nanopore sequencing reads by ~13 nt, RT-drop positions were shifted by 13 nt to accurately determine the exact RT-drop positions. To identify significant RT-drops in a given transcript, we first computed RT-drop scores at each site, which took the difference in the coverage at a given position (0) relative to the previous position (−1). We then computed the difference (Δ RT drop-off score) in RT-drop scores between CMC-probed and unprobed conditions. Lastly, we normalized the Δ RT drop-off score at each position by the median

RT drop-off per transcript, leading to final CMC scores, which can be compared across transcripts. Positions with a CMC score greater than 25 were considered significant—that is, to contain a pseudouridine. We should note that the nanoCMC-seq signal-to-noise ratio is dependent on the coverage of the individual transcript.

Demultiplexing direct RNA sequencing. Demultiplexing of the barcoded direct RNA sequencing libraries was performed using DeePlexiCon with default parameters³⁹. Reads with demultiplexing confidence scores greater than 0.95 were kept for downstream analyses. We used a lower score in the case of polysomal fractions and mRNA runs (0.80) owing to the low read coverage of some fractions and/or genes. We should note that the dataset was also analyzed using 0.95 threshold, and results and conclusions of the analysis did not change, compared to those obtained using 0.80 threshold.

Base-calling direct RNA sequencing. Reads were base-called with standalone Albacore versions 2.1.7 and 2.3.4 with the --disable_filtering parameter and standalone Guppy versions 2.3.1 and 3.0.3 with default parameters running in CPU. In-house scripts were used for computing the number of unique and common base-called reads between the different approaches as well as to compare the tendency of each base-caller regarding read lengths and qualities. Both Albacore and Guppy are available to ONT customers via their community site (<https://community.nanoporetech.com/>). Differences between the base-called features using distinct base-callers were determined using the Kruskal–Wallis test with Bonferroni correction for pairwise comparisons, whereas differences between unmodified and modified sites were assessed using the Mann–Whitney–Wilcoxon test.

Mapping algorithms and parameters. Reads were mapped using either Minimap2 (ref.⁴⁵) or GraphMap⁴⁶. Minimap2 version 2.14 was run with two different parameter settings: (1) minimap2 -ax map-ont, which is the recommended setting for direct RNA sequencing mapping, and, thus, we refer to as ‘default’, and (2) minimap2 -ax map-ont -k 5, which we refer to as ‘sensitive’. GraphMap version 0.5.2 was also run with two different parameter settings for comparison: (1) graphmap align, using ‘default’ parameters, and (2) graphmap align --rebuild-index -v 1 --double-index --mapq -1 -x sensitive -z 1 -K fastq --min-read-len 0 -A 7 -k 5, which is expected to increase the tolerance to errors that might occur under the presence of RNA modifications, and, thus, we refer to as ‘sensitive’. Yeast total RNA runs were mapped to rRNAs and ncRNA transcripts using GraphMap with default settings. Yeast poly(A)-selected runs were mapped to the yeast genome (SacCer3) using minimap2 with -ax splice -k14 -uf parameters. The scripts can be found in the GitHub repository (https://github.com/novoalab/yeast_RNA_Mod). Sequencing, base-calling and mapping statistics for all yeast sequencing runs (total RNA and poly(A)-selected RNA) can be found in Supplementary Tables 12 and 13.

Analysis of base-called features in curlcakes. SAM files were transformed into BAM files using Samtools version 1.9 (ref.⁹⁰) and were then sorted and indexed to visualize the data using Integrative Genomics Viewer (IGV) version 2.4.16 (ref.⁹¹). Base-called features were extracted with EpiNano version 1.1 (<https://github.com/enovoa/EpiNano>). PCA was used to reduce the dimensionality of the base-calling error data to visually inspect for base-calling differences, using as input the base-called features (mismatch frequency, deletion frequency and per-base quality) from all five positions of each *k*-mer. Only *k*-mers that contained a given modification once in the 5-mer were included in the analysis. All scripts used to analyze in vitro transcribed sequences using different base-calling algorithms and mappers, as well as to generate the figures related to their analysis, are available at https://github.com/novoalab/Best_Practices_dRNAseq_analysis.

Analysis of base-called features in yeast RNAs. SAM files were transformed into BAM files using Samtools version 1.9 (ref.⁹⁰) and then sorted and indexed to visualize the data using IGV version 2.4.16 (ref.⁹¹). Base-called features were extracted using EpiNano version 1.1 with minor modifications, which consisted in including in the output CSV file the directionality of mismatched bases (C_frequency, G_frequency, A_frequency and U_frequency). The modified EpiNano script can be found at https://github.com/novoalab/yeast_RNA_Mod. Scripts for the analysis and visualization of base-called features are also included in the same GitHub repository.

Visualization per-read current intensities using Nanopolish. Nanopolish event align output was processed to extract the current intensity values corresponding to the 15-mer regions centered in the modified sites for the following sites: (1) six Ψ rRNA sites for which knockout data were available (25s:2133, 25s:2129, 25s:2826, 25s:2880, 25s:2264 and 18s:1187) for all four sequencing datasets (wild-type, snR3-KO, snR34-KO and snR36-KO); (2) four Nm sites for which knockout data were available (25s:817, 25s:908, 25s:1133 and 25s:1888) for all four sequencing datasets (wild-type, snR60-KO, snR61-KO and snR62-KO); and (3) seven Ψ snRNA/snoRNA sites that were identified as heat-sensitive, for which there was a minimum of 100 reads of coverage. Reads with empty values in the 15-mer region in the Nanopolish event align output were omitted from the analysis.

Analysis of current intensity, dwell time and trace. In this work, we used two different softwares to extract current intensity: Nanopolish⁷² and Tombo⁵⁰. Nanopolish was used to extract the aligned current intensity values per read and position, using the option `--scale-events`. Mean current intensity per-position was computed by summing the current intensities of all reads aligned to the same position, divided by the total number of reads mapping at a given position. All scripts used to process Nanopolish event align output, including scripts to display mean current intensity values along transcripts, are available in GitHub (<https://github.com/novoalab/nanoRMS>).

Signal intensity, dwell time and trace were retrieved using `get_features.py` script, which is available as part of nanoRMS. This program internally uses `minimap2` (read alignment), `Tombo` (calculation of signal intensity and dwell time) and `ont-fast5-api` (retrieval of trace). Trace represents the probability that a given signal intensity chunk might be originating from each of the four canonical bases (A, C, G and T/U), and it is reported relative to the reference base. For example, in a T reference position that is incorrectly reported as C (common base-calling error observed for Ψ sites), the trace value will be reported for the reference base (T in this case). Then, the final read alignment and all the features are stored into sorted BAM files. All scripts necessary to retrieve and store per-read, per-position features and plot/calculate results are available in the nanoRMS GitHub repository (<https://github.com/novoalab/nanoRMS>).

De novo prediction of pseudouridine modifications on yeast mitochondrial rRNAs. To systematically identify Ψ sites de novo based on the Ψ base-calling signatures, we first extracted the mismatch frequency and per-base mismatch frequency (`C_freq`, `A_freq`, `U_freq` and `G_freq`) from both unmodified (U) and modified (Ψ) sites from cytosolic rRNAs, from three biological replicates. As expected, C mismatch frequency (`C_freq`) and global mismatch frequency (`mis_freq`) showed clearly distinct distributions when comparing unmodified and Ψ-modified sites (Fig. 5a). We then determined the optimal cutpoints for these two features using the `cutpointr` package in R with `oc_youden_kernel` method, which applies kernel smoothing and maximizes the Youden indexing. This approach predicted `C_freq` = 0.137 and `mis_freq` = 0.587 as optimal cutoffs. For the mitochondrial rRNA, we filtered the uridine sites based on the selected features and assigned those that are replicable in three biological replicates as 'candidate' pseudouridine sites.

De novo prediction of pseudouridine modifications in yeast mRNAs and ncRNAs. Owing to the lower stoichiometry of modification of ncRNAs (snRNAs and snoRNAs) and mRNAs, we focused on analysis of the de novo detection of Ψ sites whose pseudouridylation levels would be changing between two conditions, either by comparing normal and stress (heat-shock) conditions or by comparing the base-calling 'error' patterns of wild-type strains and *Pus1*- or *Pus4*-deficient strains. Only sites that passed the coverage filter ($n > 30$ reads) in both biological replicates from both conditions were considered in the analysis (Supplementary Table 6). Sites with minimal mismatch frequency difference of 0.1 between the two conditions in both replicates that met the identified Ψ signature (`C_freq` = 0.137 and `mis_freq` = 0.587) were considered as true Ψ sites that were heat-sensitive, *Pus1*-dependent or *Pus4*-dependent, respectively. The individual candidate Ψ mRNA and ncRNA sites identified using nanopore sequencing, as well as the previously reported Ψ mRNA and ncRNA sites (using CMC probing coupled to Illumina sequencing), can be found in Supplementary Tables 7–10.

Prediction of RNA modification stoichiometry using nanoRMS. Per-position features from individual reads were stored in BAM files using `pysam` (<https://github.com/pysam-developers/pysam>) and stored in either Numpy arrays (<https://numpy.org/>) or Pandas DataFrames (<https://pandas.pydata.org/>) using the script `get_features.py`, which is available as part of nanoRMS. Models were trained with combinations of features with diverse ranges of sequence contexts surrounding the modified sites ($k = 1-15$). Features used to predict stoichiometry included (1) current intensity (SI), (2) dwell time in the center of the pore (at position 0, DT/DT0), (3) dwell time at helicase center (shifted by 10 positions, DT10) and (4) base probability (trace, TR). Estimation of modification frequency was performed using unsupervised (GMM, KMEANS, IsolationForest and OneClassSVM) and supervised (KNN and RandomForest) machine learning methods implemented in `sklearn` (<https://sklearn.org/>). Plots were built using `matplotlib` and `seaborn` (<https://seaborn.pydata.org/>).

Trained models were first benchmarked with unmodified (KO) and modified (wild-type) reads from rRNA mutant datasets to identify which machine learning methods and which combination of features discriminated between modified and unmodified reads. Then, we tested how the diverse models would perform at diverse stoichiometries of modification. To this end, we simulated samples with varying levels of modification—0%, 20%, 40%, 60%, 80% and 100% (using mixes of KO and wild-type reads)—and estimated the modification level in those simulated samples by comparing them to KO (Supplementary Fig. 5c).

nanoRMS performed best when trained with signal intensity (SI) + trace (TR) as features and when using KNN supervised models or KMEANS unsupervised models, both for Ψ- and Nm-modified sites. Predictions by each

clustering algorithm, and for each individual rRNA modified site, are shown in Supplementary Table 4. For mRNA and ncRNA analysis, only sites with more than 30 reads of coverage in all conditions and replicates were included for predicting RNA modification stoichiometry. Prediction of RNA modification stoichiometry in mRNAs and ncRNAs was performed using signal intensity + trace as features and *k*-means as classification algorithm. Stoichiometry changes were reported as the difference in predicted stoichiometry between the two conditions. All code and examples to predict RNA modification stoichiometry are available as part of the nanoRMS GitHub repository (<https://github.com/novoalab/nanoRMS>).

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

For in vitro transcribed datasets, FAST5 files used in this work were already publicly available (UNM and m⁶A: PRJNA511582) or are publicly available in the Sequence Read Archive (SRA) (m⁶C: PRJNA563591; hm⁵C: PRJNA548268; Ψ: PRJNA549001; UNM-S: PRJNA575545; Ψ with varying stoichiometries: PRJNA695584). Base-called and demultiplexed FASTQ from all yeast RNA direct RNA sequencing data runs are publicly available in the Gene Expression Omnibus (GEO) under accession number GSE148603, including processed EpiNano outputs. FAST5 files for all yeast RNA direct RNA sequencing are available in the European Nucleotide Archive (ENA) under accession numbers PRJEB37798 and PRJEB41495. A detailed description of the datasets used and sequenced in this work, with their corresponding GEO and ENA/SRA IDs, can be found in Supplementary Table 14.

Code availability

All scripts and code used in this work are available in GitHub: https://github.com/novoalab/Best_Practices_dRNAseq_analysis (analysis of in vitro curlcake datasets), https://github.com/novoalab/yeast_RNA_Mod (analysis of in vivo datasets) and <https://github.com/novoalab/nanoRMS> (prediction of RNA modifications and estimation of RNA modification stoichiometries).

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Author contributions

O.B. and M.C.L. performed most wet lab experiments, including RNA extraction and nanopore library preparation. O.B. and L.P.P. performed bioinformatic analysis of the data, together with J.M.R. and E.M.N. O.B. conceived and performed nanoCMC-seq experiments. M.C.L. produced the in vitro transcribed sequences with modifications and their corresponding nanopore libraries. O.B. prepared and sequenced the in vitro transcribed sequences with different pseudouridine stoichiometries. L.P.P. benchmarked and wrote the nanoRMS code, together with O.B. and E.M.N. J.M.R. performed bioinformatic analyses on in vitro transcribed constructs and compared base-calling and mapping algorithms. I.M. built polysome gradients and helped with their corresponding nanopore libraries. S.C. and I.M. prepared and sequenced the 2'-O-methylation mutant strains. R.M. and H.G.S.V. cultured the *S. cerevisiae* strains under different stress conditions. H.L. contributed with code for the analysis of current intensity values. A.S.C. and S.S. cultured all snoRNA-depleted yeast mutant strains and extracted their total RNA. E.M.N. conceived the project. E.M.N. supervised the work, with the assistance

of J.S.M. M.C.L., O.B. and E.M.N. built the figures. O.B., M.C.L. and E.M.N. wrote the paper, with contributions from all authors.

Competing interests

E.M.N. has received travel and accommodation expenses to speak at Oxford Nanopore Technologies conferences. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41587-021-00915-6>.

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For in vitro transcribed datasets, FAST5 files used in this work were already publicly available (UNM and m6A: PRJNA521324), or have been made publicly available in SRA (m5C: PRJNA563591; hm5C: PRJNA548268; Ψ : PRJNA511582, UNM-S: PRJNA575545). Base-called and demultiplexed FASTQ from all yeast RNA direct RNA sequencing data runs have been made publicly available in GEO, under the accession number GSE148603, including processed EpiNano outputs and Nanopolish outputs. FAST5 files for all yeast RNA direct RNA sequencing are available in ENA under accession PRJEB37798 (total RNA) and PRJEB41495 (mRNA). A detailed description of the datasets used and sequenced in this work, with their corresponding GEO and ENA/SRA ids can be found in Table S14.

Data analysis

All scripts and code used in this work have been made available in GitHub: https://github.com/novoalab/Best_Practices (analysis of in vitro curlcake datasets), https://github.com/novoalab/yeast_RNA_Mod (analysis of in vivo datasets) and <https://github.com/novoalab/nanoRMS> (prediction of RNA modifications and modification stoichiometries).

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in SRA (m5C:PRJNA563591; hm5C: PRJNA548268; Ψ:PRJNA511582, UNM-S: PRJNA575545). Base-called and demultiplexed FASTQ from all yeast RNA direct RNA sequencing data runs have been made publicly available in GEO, under the accession number GSE148603, including processed EpiNano outputs and Nanopolish outputs. FAST5 files for all yeast RNA direct RNA sequencing are available in ENA under accession PRJEB37798 (total RNA) and PRJEB41495 (mRNA). A detailed description of the datasets used and sequenced in this work, with their corresponding GEO and ENA/SRA ids can be found in Table S14.

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